Media Wall & Dining Wall: Build Guide

Version 23 · Oct 20, 2026 · for Emma and Dave · sizes in cm; every step carries its own drawings, pictures and cut list (the full plan has the fit proof)



This guide assumes you've hung shelves and put flat-pack together, but haven't cut a joint before. Every step says what to do, what to check before moving on, and roughly how long it takes the first time. Each step carries a coloured label for its part of the house (Prep, Network, Power & cables, Paint, Media wall, Dining table, Lighting, AV), and so does each phase of the buy list. You're doing the electrics yourselves: steps 3 and 12 have the safe routine and the point where you stop and call an electrician. Sizes are finished sizes in centimetres.

Before you start: five rules

- **Check every wall before you drill.** Run the cable and pipe detector over the spot, and never drill straight above, below or beside a socket or switch: cables run vertically and horizontally from them.
- **Measure twice, mark once, cut once.** Walls are rarely square: measure the alcove at the floor, 76, 134, 155, 196 and the ceiling, and cut to the smallest.
- **Two people for anything over 15 kg**: the uprights, the worktop, the table top, the TV.
- **The electrics are yours, within limits.** Swapping the socket for a double and the two ceiling fittings are like-for-like jobs, not notifiable under Part P in England, as long as they meet the wiring regulations. Isolate, lock off and prove dead every time. Anything new (a new circuit, the consumer unit, work by a bath or shower) is an electrician's, and so is anything on the stop list in section 9.
- **Dry-fit everything** (assemble without glue or screws) before the final fix.

Tools

You don't need a workshop. Hire the two big tools for a weekend (a track saw and an SDS drill), or ask the timber merchant to cut the oak and ply to the cut lists: most will, for a small charge, and that removes the hardest cuts.

Tool	What it's for	Get it	About £
MEASURE AND MARK			
Tape measure (5 m) and a steel rule	every cut and every height	Buy	10–15
Combination square	squaring lines across boards and checking corners	Buy	10–15

Spirit level, 60 cm and 120 cm	levelling the base, the shelves and the table	Buy	15–30
Cross-line laser level	marking all the heights round the alcove in one go	Buy or borrow	30–60
Cable and pipe detector	checking every wall before you drill: non-negotiable	Buy	15–30
Pencil, masking tape, a marking knife	marking cuts (tape stops oak splintering)	Buy	5
CUT			
Track saw, or a circular saw with a straight-edge guide	the worktop, the table top and the long shelves	Hire or borrow	25–40 a day
Jigsaw	notches and the cable slot	Buy or borrow	40–70
Tenon saw (Japanese pull saw is easiest)	the tenons and the wedge kerfs	Buy	15–25
Sharp chisels, 6, 12 and 25 mm, and a mallet	cleaning out mortises and housings	Buy	30–50
DRILL AND FIX			
Cordless drill-driver (18 V)	pilot holes and screws; most of the job	Buy or borrow	60–120
SDS or hammer drill with masonry bits	plugs into the brick walls and the chimney breast	Hire or borrow	15–25 a day
Dowel jig and 10 mm brad-point bit	the dowels under the uprights and at the wrap ledge	Buy	20–35
Clamps: four 60 cm sash or F-clamps, four small	glue-ups and holding work while you fix it	Buy	40–60
Stud finder (if any wall is plasterboard)	finding studs to fix into	Buy	15
SHAPE AND FINISH			
Router (1/4 in is enough) with a straight bit and a guide	the housings, the LED grooves and the cable groove	Hire, borrow or buy	60–120
Random-orbit sander, 120 and 180 grit	smoothing the oak before oiling	Buy or borrow	40–70
Hardwax oil, lint-free cloths, a brush	finishing the oak	Buy	25–35
ELECTRICS (YOUR OWN: STEPS 3 AND 12)			
Two-pole voltage tester to GS38, with its proving unit	proving every wire dead before you touch it: non-negotiable	Buy	35–70
MCB lock-off and a small padlock	keeping the circuit off while you work on it	Buy	8–15
Plug-in socket tester	checking the new double socket is wired the right way round	Buy	8–15
Insulated (VDE) screwdrivers, wire strippers, side cutters	the socket and the ceiling fittings	Buy	20–40
Wall chaser with dust extraction (or a bolster and club hammer)	the PC's two conduits in the side wall	Hire	25–45 a day
Cat6 punch-down tool and a cable tester	terminating and checking the Cat6 runs	Buy	20–35
SAFETY			
Safety glasses, ear defenders, FFP2 dust masks	every cut, every sanding job; oak dust is an irritant	Buy	15–25

Work gloves; a first-aid kit	handling boards and sheet goods	Buy	10
A sturdy step ladder	the high shelf, the ceiling and the PC shelf	Borrow	—

Six techniques you'll use

1. **Fixing into brick.** Detect, mark, drill with the SDS on hammer setting, clear the dust, push in a brown (7 mm) wall plug and drive a 5 × 60 mm screw. In plasterboard, find a stud, or use heavy-duty plasterboard fixings rated for the load.
2. **Cutting oak cleanly.** Put masking tape over the cut line, score it with a knife, cut with the good face down (circular or track saw) or up (jigsaw). Clamp a straight edge as a guide.
3. **Routing a housing** (a shallow slot a shelf sits in). Clamp a straight batten as a guide, set the bit 4 mm deep, rout, then step down to 12 mm in passes. Practise on an offcut first.
4. **A wedged through-tenon.** Mark the 6 × 2.4 cm mortise on both faces of the upright and drill most of it out with a 20 mm bit from each side. Chisel it to the line. Saw the tenon on the shelf's end. Saw a kerf down its middle, glue it, tap it home, then drive a glued oak wedge into the kerf. When it's dry, saw it flush and plane it.
5. **Dowelling.** Use a dowel jig to drill matching 10 mm holes 3 cm deep in both parts, glue the dowels, clamp.
6. **Oiling.** Sand to 180 grit, wipe off the dust, rub on a thin coat of hardwax oil, wipe off the excess after 15 minutes, let it dry overnight, give it a second coat. Do the undersides too, so the boards don't cup.

The order of work

#	Stage	What to do	Check before moving on	Time
1	PREP **Measure and count**	The alcove's width at the floor, 76, 134, 155 and 196 cm and at the ceiling: cut everything to the smallest. The floor's level across the alcove, and the chimney breast's flatness with a long level. The door's swing, and the sofa: overall width, depth with the return, back height. Count the LPs and games (PS2, Xbox 360, Xbox One), and measure ten cases of each. Make a cardboard template of the alcove at worktop height: walls are rarely square.	Every figure within 1 cm of the plan. If not, send the numbers and the drawings are redone first.	2 h
2	NETWORK **Network, stage 1**	Unplug everything from the Hub 5. Log in at 192.168.0.1 with the password on its base, then Modem Mode → Enable Modem Mode → Apply. Its page moves to 192.168.100.1. Connect a hub port to Deco A in bedroom 3 and set it up in the Deco app, reusing the old Wi-Fi name and password. If it shows no internet, try another hub port and restart the hub. Plug the new 8-port Switch 1 into Deco A's other port, then the PCs into Switch 1. Add Deco B in the living room and Deco C in the kitchen. For now they link over Wi-Fi. Speed-test each room, and check the Xbox's network settings show NAT: Open.	Internet in every room, and NAT: Open on the Xbox.	2 h

3	POWER & CABLES NETWORK **Cables and the socket, your own job**	The Cat6: two runs from bedroom 3, across the landing and down through the airing cupboard, one to the alcove's right corner for Deco B, one on to the kitchen for Deco C, a spare pulled with each; the optional runs to the master and bedroom 2 if you want them. Low voltage: no rules beyond care. The socket: swap the single on the side wall for a double at the same spot, with the circuit isolated, locked off and proved dead (the safe routine is in this step). Like for like, so not notifiable. The PC's two conduits: chase them into the solid side wall from behind the PC shelf to skirting height and behind the skirting to the base, 38 mm oval for the HDMI and Cat6, 25 mm for the PC's mains lead, a draw wire in each. Make good. Stop and call an electrician if anything on the stop list turns up (old wiring, no earth, a reading where there should be none, anything you can't match). Before you fill a chase, photograph it with a tape measure in shot, and mark every route on the plan.	The socket tester shows correct wiring on both outlets, both Cat6 runs test good end to end, and every route is photographed.	2–3 days
4	PAINT **Paint**	Paint an A4 board in Card Room Green and look at it by day and with the LED strip on. Wash the skirting and door with sugar soap, sand to a key, and prime them. Two coats of emulsion on the walls, ceiling, alcove and chimney breast. Two coats of eggshell in the same green on the skirting and radiator (the door and its frame stay white). Leave it a week to cure before fixing anything to it.	Two even coats, and no white showing at the edges or behind the radiator.	2–3 days, then a week to cure
5	PREP **Mark the heights**	With the laser level: 73 and 76 for the worktop, 130 and 134 for the Series X shelf, 153 and 155.4 for the ledges, 192 and 196 for the high shelf. The uprights at 0–4.5 and 131–135.5 cm from the side wall. The dining wall's backboard at 51.3–73.3, centred on the breast. Run the cable detector along every line you will drill, and mark anything it finds.	Every line level end to end, and the uprights' lines plumb.	1 h
6	MEDIA WALL **Build the base**	Order the oak and ply from the cutting list (below the steps), a week or two ahead: most merchants cut to size for a small charge. Build the 9 cm plinth for bays 1–2, with the open bay for the hidden drawer under bay 2. Level it with shims. Assemble the two BESTÅ frames with their back panels left off and stand them on the plinth. Build bay 3 from 18 mm ply, 60 × 40 × 73, and bolt it to bay 2. Screw the 4 cm spacer blocks to the wall, push the base against them and fit the anti-tip brackets. Lay the flat two-compartment trunking in the channel behind (mains one side, data the other) and run the cables in.	The tops level within 1 mm across 186 cm, and nothing rocks.	1 day
7	MEDIA WALL **Fit the worktop**	Cut the worktop and the dining wall's backboard from the one 300 × 62 oak board, using the template from step 1. Scribe the ends to the walls; cut the TV slot over the channel and the grommet hole behind the deck. Oil it on every face, undersides and cut ends included. Screw it up through the frame tops in slotted holes, so the oak can move.	Tight to both walls, and the slot sits over the channel.	½ day
8	MEDIA WALL **Fit out the base**	Shelves on their pins at the fit-proof heights for the consoles. The game drawer in bay 1: runners on the frame's sides, then the box (its cut list is with this step). Bay 3: the 12 mm divider first (17.8 cm in from the left side), then the LP crates' runners at 2.3 and 37.0 cm on the right and the three slim drawers' at 2.3, 25.1 and 47.9 on the left, then each box. The hidden plinth drawer under bay 2. Bay 2: the divider and the part shelf, then the sub on its riser, its front flush with the face frame.	Every drawer runs smoothly, the two crates clear each other by 2 cm, and the slim drawers by 2.8.	1–2 days

9	MEDIA WALL **Uprights and shelves**	On the bench: cut the housings and the mortises in both uprights, and the LED groove and cable groove. Dry-fit each upright with its shelves before any glue. Lay Deco B's leads in the groove behind the right upright. Stand the uprights on their dowels and fix the concealed brackets to the back wall. Fit the high shelf and the ledge, then the Series X shelf: wedge its tenons, and peg the housings. On the chimney breast: the “now playing” ledge, its top at 104, on two concealed brackets, 6 cm in from each side of the breast (with the table by the stairs).	Uprights plumb, shelves level, the TV slides in with 2 cm each side, and the “now playing” ledge is level with an LP standing on it.	2–3 days
10	MEDIA WALL **The side-wall run and the PC end**	Cut the run from its 225 × 30 × 4 board: mark the front edge from the full-size template (7 cm deep along the LPs, sweeping out to 28 under the PC, a 5 cm radius on the end), jigsaw just outside the line and rout to it with a bearing-guided bit; rout the LED groove under the front edge. Screw the oak cleat along the side wall and fit the concealed brackets for the PC end. Lift the run on, dowel its end to the left upright's front edge, and screw it down to the cleat from above (plugged). Screw the keyboard tray's housing under it and fit its runners. Fit the brush grommet and pull the PC's leads down the two chased conduits with the draw wires: data in the big one, its mains lead in the small one. The PC's extras (v17): screw the powered USB hub under the run's front edge below the PC, fit the remote power button's card and stick its button under the wrap ledge by the sofa, and lay the magnetic filter on the top exhaust.	One level line with the ledge over the TV, the sweep runs smooth with no flat spots, and the PC end holds your weight leaning on it.	1 day
11	DINING TABLE **The flip-top dining table**	Screw the backboard to the wall, plugs every 20 cm, and glue and screw the 7.3 cm hinge rail along its top. Fit the two Hebgo brackets on its face, 80 cm apart, arm tops level with the rail's top. On the bench: rout the three flip-top hinges into the meeting edges of the top and the half, and fit them; hinge the gate leg under the half, 3 cm in from its far edge, with its two stays. Screw the lift-off hinges to the rail's front edge and the top's back edge, all the same hand, and hang the pair. Fit the two turn buttons and the magnetic catch above it.	It opens onto both brackets with a click each side, the half flips out flush and its gate leg locks upright; it folds back flat and swings up to the wall without binding, and nothing moves when you lean on it open.	1–2 days
12	LIGHTING **Lighting: smart spot bars and the LED strips, your own job**	Isolate the lighting circuit at the consumer unit, lock it off and prove it dead at each ceiling point. Take down the two old fittings. The light switches stay as they are: the smart bulbs need full power, so no dimmers. Hang bar 1 on the rose between the TV and the sofa and bar 2 on the pendant point by the stairs, earth first, both running from the TV towards the window; fit the smart colour bulbs and point every head straight down, evenly spread; tilt one only if its pool lands badly. The LED strips in the uprights' grooves and under the side-wall shelf on their plug-in driver (on the 10-gang), the ceiling strip through its notches, the bias light on the TV; your red lamp in its spot. Power back on, switches on: add the eight bulbs to the app, group them as bar 1 and bar 2, and save the five lighting scenes (Paint and light), General first.	Every bulb answers the app, the scenes recall, and from the sofa no spot head shows in the screen or shines in your eyes.	1 day

13	AV NETWORK **Connect and finish**	TV inputs: HDMI 1 the PC, HDMI 2 the soundbar (eARC), HDMI 3 the Series X, HDMI 4 the HDMI switch (Xbox One, 360, PS3, RetroTINK). The PS2 by component into the RetroTINK and by optical to the soundbar; the PS1 by RGB SCART. The deck to the preamp, the preamp to the soundbar's AUX; set the tracking weight; soundbar in Music mode for vinyl. Network stage 2: plug each Cat6 run into its Deco, check Deco B and C show "Ethernet", and give the Series X and PC high priority. Label both outlets of the double socket and each HDMI switch input; set the sub and rear levels from the sofa.	Everything on the network's "done when" list, and every source plays through the soundbar.	1 day

Roughly 14–18 working days spread over a few weekends, the electrics and the Cat6 included, plus the paint drying. Steps 6–11 can be split: the base and dining wall first, the oak joinery once you're confident.

Where beginners usually go wrong

- **Cutting the worktop to the tape instead of to a template.** Make a cardboard template of the alcove first; walls are rarely square.
- **Skipping the dry-fit.** Gluing a tenon that doesn't fit is the one mistake you can't undo.
- **Oil on one side only.** The board cups within a week.
- **Loading the LP crates before the anti-tip brackets go in.** A full crate weighs about 30 kg.
- **Folding the table up with something on it.** Clear it first, every time, and keep both turn buttons turned when it's up.

The cutting list

Every made part in one list, grouped by what it's cut from, for the timber merchant. Sizes are finished sizes in cm, length × width × thickness. The cut lists in the other sections say how each part joins; this list is for ordering.

Solid oak, hardwax oil

Ask for it planed all round; scribe the uprights and worktop on site

Part	Size (cm)	Qty	Notes	Section
Uprights	159 × 30 × 4.5	2	Buy 165 long and scribe to the ceiling; housings, mortises, LED and cable grooves cut on the bench	Media wall
High shelf, long length	128.9 × 30 × 4	1	Housed 1.2 into both uprights and pegged	Media wall
High shelf, short length	51.7 × 30 × 4	1	Housed into the right upright; cleat on the reveal	Media wall

Series X shelf	55 × 30 × 4	1	Two through-tenons 6 × 2.4, wedged flush	Media wall
Record ledge	128.9 × 7 × 2.4	1	Housed and pegged	Media wall
Side-wall run (v15)	220 long × 7 sweeping to 28 × 4, from one 225 × 30 × 4 board	1	On an oak cleat, concealed brackets under the PC end, dowelled to the left upright; LED groove under the front edge	Media wall
Ledge lips	128.9 × 1.5 × 0.9 · 126 × 1.5 × 0.9	1 · 1	Glued along the ledge's and the run's front edges	Media wall
Keyboard tray housing	70 × 28 × 7, sides 7 × 2	1	Screwed under the PC shelf	Media wall
Worktop	186 × 46 × 3	1	From one 300 × 62 full-stave worktop; chamfered front	Media wall
Dining backboard · hinge rail	110 × 22 × 3 · 110 × 4 × 7.3	1 · 1	From the same worktop's offcut	Dining table
Face frame	186 × 73 overall, 2 thick, stiles about 5 wide	1 set	End stiles scribed to the walls	Media wall
Table top and the half, centres	101 × 80 × 2.7	2	Edge-glued staves: 96 between the shoulders, plus a 2.5 tongue each end (v15: the half matches the top)	Dining table

Breadboard ends	80 × 7 × 2.7	4	Pegged with three 10 mm pegs each, in slotted holes	Dining table
Gate legs	73.3 × 4.5, tapering to 3	2	Tapered on the two inside faces; tenoned into the apron (v15)	Dining table
Gate apron	94 × 7 × 2	1	Between the legs; two butt hinges along its top edge, under the half	Dining table
Turn buttons	6 × 2.5 × 2.2	2	On a single screw each	Dining table
Drawer fronts	game drawer 56.8 × 6.5 × 2 · plinth drawer 54 × 6 × 2	1 · 1	Finger pull cut underneath	Media wall
LP crate lips · slim drawer lips	36.3 × 5 × 1.5 · 16.5 × 5 × 1.5	2 · 3	Cut to the face frame's openings, less 3 mm (v16)	Media wall
“Now playing” ledge	105.5 × 7 × 2.4, lip 1.5	1	On the breast at 104, two concealed brackets (v16; with the table by the stairs)	Media wall
Follower blocks · Xbox One lift block	19 × 12 × 2 · 15 × 12 × 2	2 · 1	Offcuts, felted underneath	Media wall
Wedges · pegs · dowels	6 × 3, tapering · 10 mm dowel, 5 long · 10 mm dowel	2 + spares · 6 + 6 · 8	Glued	Media wall
Chair cupboard sides	189 × 20.2 × 1.8	2	The one against the pipe boxing is scribed to it; a 3 cm plinth notch at the front (v20)	Dining table

Chair cupboard top · floor · shelf	47.4 × 19.6 × 1.8	3	Housed into the sides; the shelf at 93, its 3 cm lip and the floor's stop the chairs tipping forward (v20)	Dining table
Chair cupboard doors	51 × 92.7 · 51 × 95.7 × 1.8	1 · 1	Two doors, lower and upper, hinged on the table's side; plain oiled oak like the folded table (v20)	Dining table

Birch ply, 18 mm

Part	Size (cm)	Qty	Notes	Section
Bay 3 sides	73 × 40	2	The record carcass, standing on the floor	Media wall
Bay 3 top and bottom	56.4 × 40	2	Between the sides; back left open to the channel	Media wall
Plinth rails	120 × 9	2	Front and back of the ladder frame under bays 1–2	Media wall
Plinth cross-pieces	34.4 × 9	3	At both ends and under the bay 1–bay 2 joint, so the bay under bay 2 stays clear for the drawer	Media wall

Birch ply, 12 mm

Part	Size (cm)	Qty	Notes	Section
Game drawer sides	35 × 15.5	2		Media wall
Game drawer back · front panel	51.9 × 15.5 · 51.9 × 6	1 · 1		Media wall
LP crate bases · slim drawer bases	34.9 × 40 · 15.3 × 40	2 · 3	The sides and ends stand on them (v16)	Media wall
Bay 3 divider	69.4 × 38	1	17.8 in from the left side, the full height inside (v16)	Media wall
Plinth drawer sides · ends	34 × 4.4 · 49.6 × 4.4	2 · 2	A 52 × 34 × 5 box	Media wall
Bay 2 divider · part shelf	35.3 × 38 · 32.2 × 36	1 · 1	The sub's column on the left; the PS2 over the PS1 on the right	Media wall
Keyboard tray	54 × 26	1	Oak lip on its front edge	Media wall

Birch ply, 9 mm

Part	Size (cm)	Qty	Notes	Section

Game drawer base	52.5 × 34.4	1	In a 6 mm groove	Media wall
LP crate sides · ends	40 × 8 · 33.1 × 8	4 · 4	Two of each per crate: a 33 cm trough (v16)	Media wall

Birch ply, 6 mm

Part	Size (cm)	Qty	Notes	Section
Game drawer cross divider	51.9 × 9	1	13 cm back from the front	Media wall
Game drawer bin walls · bin floors	12.6 × 9 · 16 × 12.6	2 · 2	Floors raked so the cases lean about 13°	Media wall
Plinth drawer base	52 × 34	1	Screwed under the box	Media wall
Slim drawer sides · backs and fronts	40 × 10 · 14.1 × 10	6 · 6	Three boxes 15.3 wide, 14.1 inside (v16)	Media wall

Hardware

Runners are sold in pairs

Part	Size (cm)	Qty	Notes	Section
LP crate runners · slim drawer runners	400 mm full extension, 68–70 kg a pair · 400 mm light duty	2 pairs · 3 pairs	Into bay 3's sides and the divider (v16)	Media wall
Game drawer runners	350 mm full extension, soft-close	1 pair	Accuride DZ3832EC or Screwfix 964VN	Media wall
Plinth drawer and keyboard tray runners	300 mm full extension · 300 mm light duty	1 pair · 1 pair		Media wall
Häfele Hebgo folding brackets	287.43.428, 580 arm	2	500 kg a pair	Dining table
Lift-off hinges	Union PowerLoad 100 × 88, all the same hand	3		Dining table
Flip-top hinges	Lee Valley 40574, solid brass, 1.27 × 7.0	3	Routed into the meeting edges of the top and the half	Dining table
Gate hinges · folding stays	75 mm steel butt hinges · locking stays	2 · 2	The gate swings down and locks; folds flat inside the half	Dining table
Sub isolation riser	dense foam or sorbothane, about 22 × 28 × 3	1	Under the sub in bay 2	Media wall
Concealed brackets	for the uprights; two for the PC shelf, 40 kg+ a pair	1 set		Media wall

Step by step, in full

Step 1 · Measure and count · You · 2 h

PREP

Tools

- Tape measure
- Spirit level, 120 cm
- A notebook (or the questions list on the plan page)



The whole room, from the window.

1. The alcove's width at the floor, 76, 134, 155 and 196 cm and at the ceiling: cut everything to the smallest.
2. The floor's level across the alcove, and the chimney breast's flatness with a long level.
3. The door's swing, and the sofa: overall width, depth with the return, back height.
4. Count the LPs and games (PS2, Xbox 360, Xbox One), and measure ten cases of each.
5. Make a cardboard template of the alcove at worktop height: walls are rarely square.

Check before moving on: Every figure within 1 cm of the plan. If not, send the numbers and the drawings are redone first.

Step 2 · Network, stage 1 · You · 2 h

NETWORK

Buy first: 1 Network, now (the buy list).

Parts

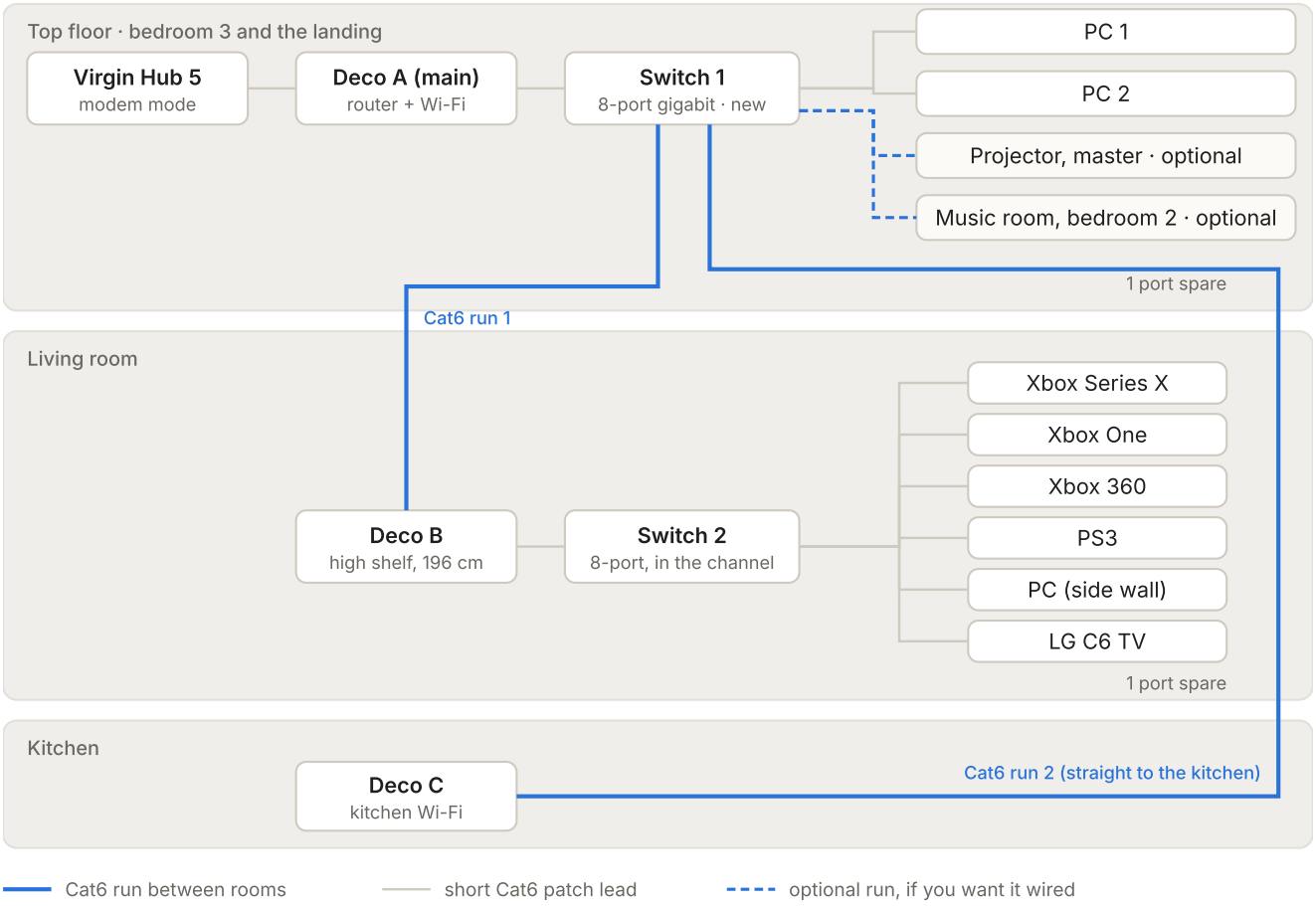
- Two TP-Link TL-SG108 8-port switches
- Cat6 patch leads, 0.5–2 m (11)

Tools

- A phone with the Deco app
- A laptop for the hub's settings page

Every floor has its own Cat6 link back to bedroom 3

Deco A and Switch 1 live with the Hub 5 and the gaming PCs in bedroom 3; two spare ports can wire the projector and the music room



1. Unplug everything from the Hub 5. Log in at 192.168.0.1 with the password on its base, then Modem Mode → Enable Modem Mode → Apply. Its page moves to 192.168.100.1.
2. Connect a hub port to Deco A in bedroom 3 and set it up in the Deco app, reusing the old Wi-Fi name and password. If it shows no internet, try another hub port and restart the hub.
3. Plug the new 8-port Switch 1 into Deco A's other port, then the PCs into Switch 1.
4. Add Deco B in the living room and Deco C in the kitchen. For now they link over Wi-Fi.
5. Speed-test each room, and check the Xbox's network settings show NAT: Open.

Check before moving on: Internet in every room, and NAT: Open on the Xbox.

Step 3 · Cables and the socket, your own job · You + a helper · 2–3 days

POWER & CABLES NETWORK

Buy first: 2 Cables and power (the buy list).

Parts

- Cat6 solid pure copper (not CCA), 2–4 runs
- RJ45 keystones and two surface boxes, or RJ45 plugs
- A double socket in the same finish, and a deeper back box if the old one is shallow
- Conduit: 38 mm and 25 mm oval, about 4 m each, two draw wires
- A 5 m C13 lead with a rewireable 13 A plug, for the PC
- One-coat filler

Hardware and fixings

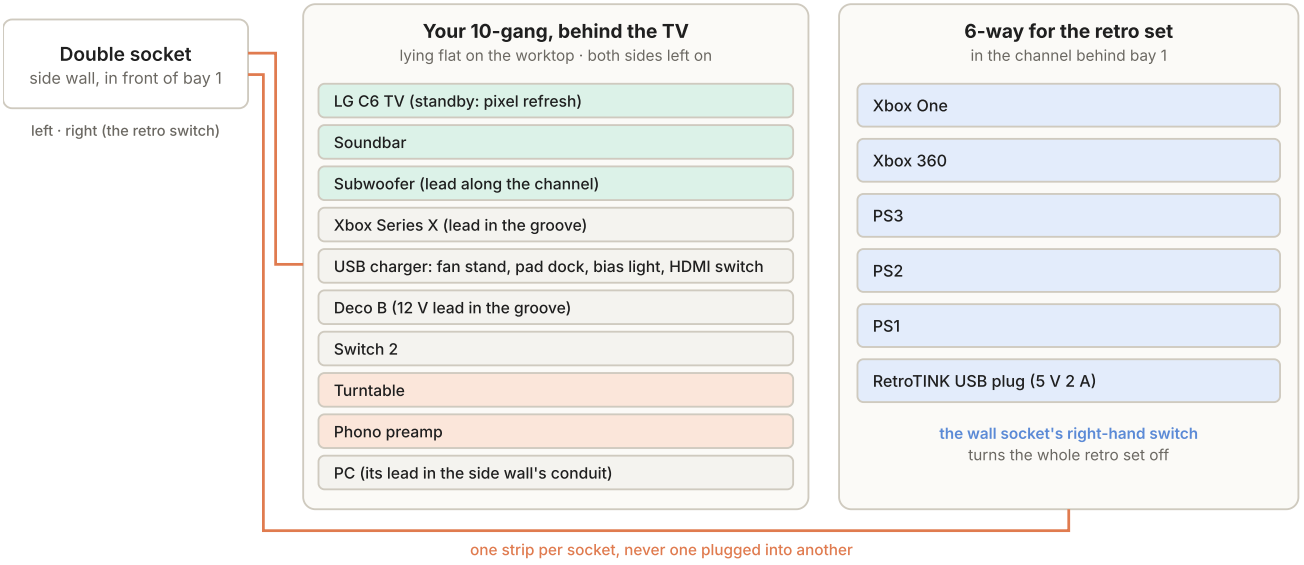
- Cable clips for the Cat6 in the loft space or under the boards
- Brown wall plugs and 4 × 30 screws for the conduit clips

Tools

- Two-pole voltage tester (GS38) and its proving unit
- MCB lock-off and padlock
- Plug-in socket tester
- Insulated screwdrivers, wire strippers, side cutters
- Cat6 punch-down tool (or crimper) and a cable tester
- Draw rods
- Wall chaser with dust extraction, or a bolster and club hammer
- Cable and pipe detector
- SDS drill
- Safety glasses, ear defenders, FFP2 masks
- A helper

Power map: your 10-gang (now with the PC) plus a 6-way for the retro set

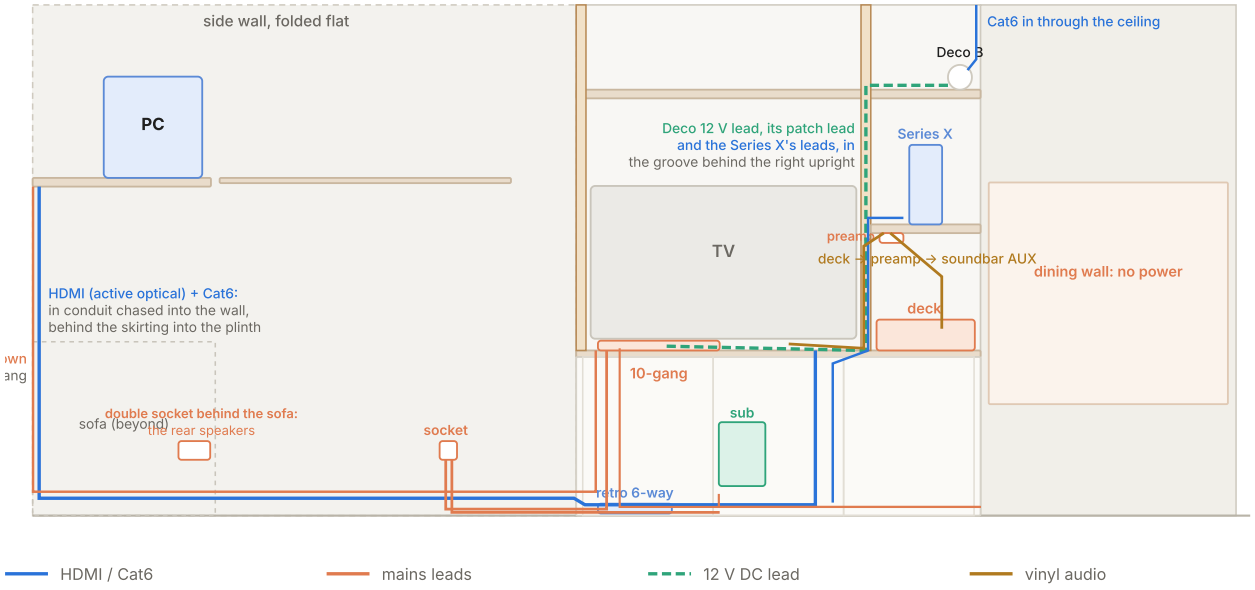
Everything modern on the left outlet, always on; the five retro consoles and the RetroTINK on the right, switched at the wall



- The socket on the side wall is a single today: you swap it for a double at the same spot (like for like, your own job: step 3).
- The retro switch moves to the wall: with the TV 2 cm from the left upright, the 10-gang's own switches can't be reached any more.
- Rear speakers: your double socket behind the sofa's short side (v23). Your lamp and the grow lights: the double behind the plant stand.
- The sub plugs into the 10-gang: its lead drops through the worktop's slot to the back of bay 2. Both spot bars use existing ceiling points.
- Load check: the 10-gang about 1.55 kW with everything on, PC included (850 W at most); the 6-way up to 0.8 kW. Each under 13 A (3 kW).
- Check the 10-gang is surge-protected; if not, use a surge-protected one for the TV and consoles. Label both outlets.

Cable routes: everything ends in the channel behind the base

Front view, schematic, with the side wall folded flat to the left so the PC's route shows; the sub sits in bay 2



Before you start: the stop list

1. Stop and call an electrician if: the tester shows anything live after you've isolated; the wiring is old (rubber or fabric sheath, or red and black with no earth); there are more wires than you can match to your photo; there are scorch marks or the old socket is loose or hot; the consumer unit has rewireable fuses or no RCD; or the job grows into a new circuit.
2. Nothing in this step is notifiable in England: the socket is a like-for-like swap on an existing circuit, outside a bathroom. It must still be done to the wiring regulations, which is what this routine is for.

The Cat6 runs (low voltage: no isolation needed)

3. Measure each route on the house plan and add 2–3 m: bedroom 3, across the landing, down through the airing cupboard.
4. Lift a floorboard in the airing cupboard. Drill a 20 mm hole through the plate below (detector first), and feed a draw rod down to the alcove's right corner. Tape both Cat6 cables and a spare to it and pull them through. Pull the kitchen run on down the same way, past the right upright's groove.
5. Keep the Cat6 away from mains: cross at right angles, never run alongside in the same hole. Clip it every 30 cm; never staple through it or kink it.
6. Punch each end down onto a keystone in a surface box (or crimp a plug), to the T568B colours on both ends. Label both ends.
7. Test each run with the cable tester: all eight lights in order. Re-terminate any that don't.

Isolate the socket circuit

8. Tell everyone in the house you're working on the electrics.
9. At the consumer unit, switch off the breaker for the side-wall socket's circuit (usually a ring marked "sockets"). If the labels aren't clear, switch off the main switch instead.
10. Fit the MCB lock-off and padlock to that breaker, keep the key in your pocket, and tape a note on the board: "Working on this circuit, do not switch on".
11. Prove your voltage tester on its proving unit: it must light.
12. At the socket, with it pulled forward on its cables, test between every pair: live and neutral, live and earth, neutral and earth. No reading on any of them.
13. Prove the tester on the proving unit again. Only now is it safe to touch.

Swap the single socket for a double

14. Unscrew the old socket and ease it out. Photograph the wiring before you undo anything.

15. Check the stop list against what you see: an earth (green-and-yellow) wire present and sleeved, no scorching, the cable sheath into the box.
16. If the double needs a bigger box, chop the plaster out round the old box to the new box's outline and fit the new box with plugs and screws, the cables through its grommets.
17. Connect wire for wire: brown (or red) to L, blue (or black) to N, green-and-yellow to the earth terminal. Two cables in the old box go into the same terminals together. Strip only enough so no copper shows outside a terminal, and tighten every screw firmly.
18. Fold the cables in neatly without trapping them, and screw the socket on level.
19. Remove the lock-off, switch on, and plug the socket tester into each outlet: it should show correct wiring on both.

The PC's two conduits in the side wall

20. Mark the route from behind the PC shelf's back corner straight down to skirting height, then along behind the skirting to the base's plinth. Run the detector along it: move it if anything shows. Never chase near the socket's cables, which run straight up, down or across from it.
21. Chase two channels side by side, 5 cm apart: one for the 38 mm oval conduit (data: HDMI and Cat6), one for the 25 mm (the PC's mains lead). Vertical only on the wall, no deeper than the conduit needs. Wear the mask: the dust is silica.
22. Take the skirting off along the route, fix both conduits with a draw wire inside each, and refit the skirting over them.
23. Fill the chase with one-coat filler, sand flush, and photograph the route with a tape measure in the shot before you paint.

Finish

24. Mark every route on the plan, and keep the photos with it.

Check before moving on: The socket tester shows correct wiring on both outlets, both Cat6 runs test good end to end, and every route is photographed.

Step 4 · Paint · You · 2–3 days, then a week to cure

PAINT

Buy first: 3 Paint (the buy list).

Parts

- Card Room Green emulsion and eggshell, primer and undercoat
- Sugar soap, sanding pads, woodwork primer

Tools

- Roller, tray and extension pole
- 25 and 50 mm brushes
- Dust sheets, masking tape
- Step ladder

1. Paint an A4 board in Card Room Green and look at it by day and with the LED strip on.
2. Wash the skirting and door with sugar soap, sand to a key, and prime them.
3. Two coats of emulsion on the walls, ceiling, alcove and chimney breast.
4. Two coats of eggshell in the same green on the skirting and radiator (the door and its frame stay white).
5. Leave it a week to cure before fixing anything to it.

Check before moving on: Two even coats, and no white showing at the edges or behind the radiator.

Step 5 · Mark the heights · You · 1 h

PREP

Tools

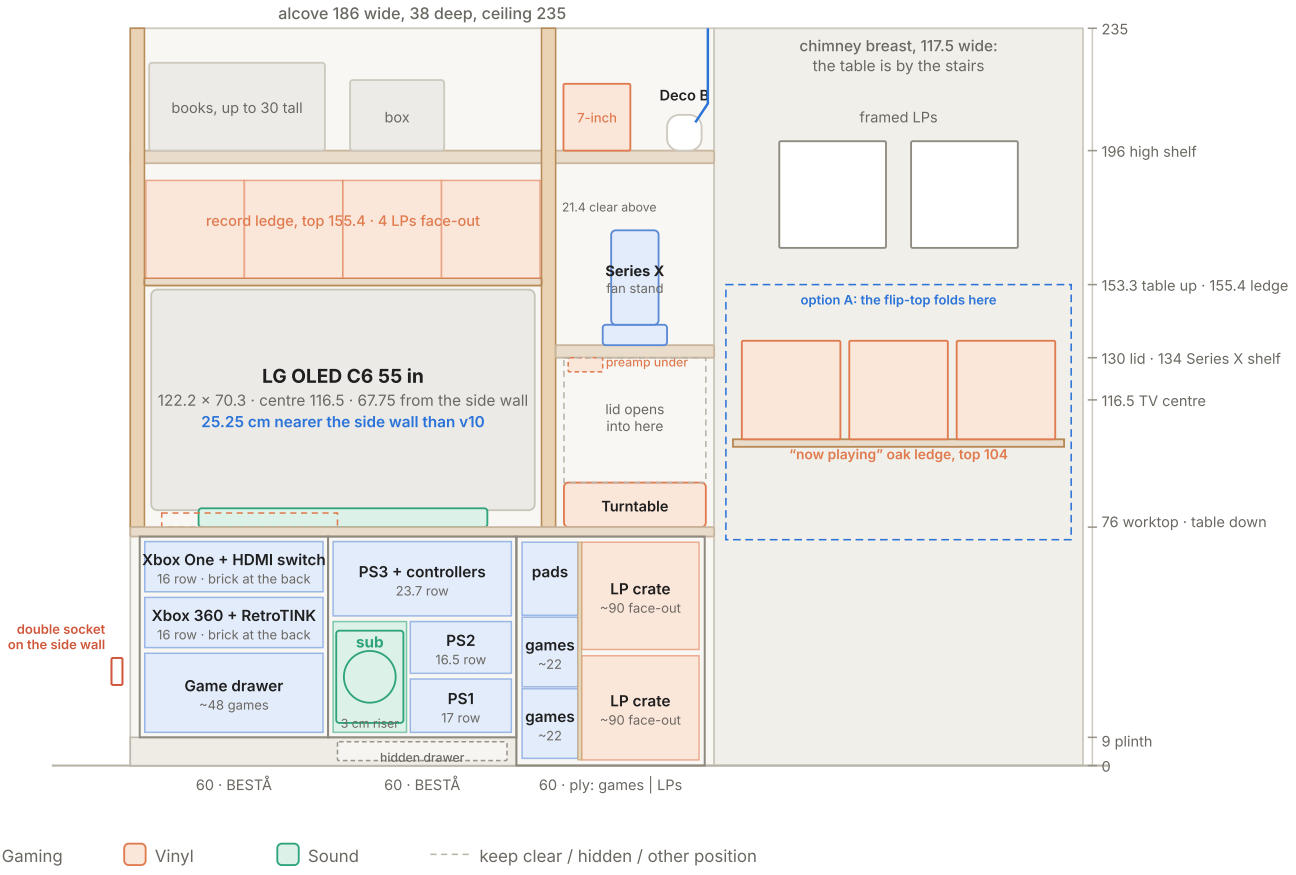
- Cross-line laser level
- Tape measure
- Pencil and masking tape
- Cable and pipe detector
- Spirit level, 120 cm



The media wall with its parts numbered.

The media wall and the chimney breast from the front, to scale

The TV between oak uprights nearer the side wall, the turntable zone on its right, three bays below (bay 3: games beside the LP crates); the “now playing” ledg



Set up a datum

1. Find the floor's highest point across the alcove (you measured its level in step 1). Every height in this plan is measured from there.
2. Stand the laser in the middle of the room, level it, and mark a line at exactly 100 cm above that high point on the alcove's back wall, both side walls and the chimney breast.
3. Measure every other height up or down from the 100 line, not up from the floor: the floor is the thing that isn't level.

The alcove's heights

4. Put a strip of masking tape down each side wall and the back wall of the alcove and mark these heights on it:
5. 73: the tops of the frames and bay 3 (the worktop's underside).
6. 76: the worktop's top.
7. 130 and 134: the Series X shelf, underside and top.
8. 153 and 155.4: the record ledge and the wrap ledge, underside and top.
9. 192 and 196: the high shelf, underside and top.
10. Carry the 153 and 155.4 lines along the side wall to 250 cm out from the alcove's back wall, for the wrap ledge and the PC shelf.

Across the back wall

11. Mark the uprights' positions from the side wall: the left one 0–4.5, the right one 131–135.5. Turn the laser to its vertical line and draw both lines from the worktop line to the ceiling.
12. Mark the base from the side wall along the floor: filler 0–3, bay 1 3–63, bay 2 63–123, bay 3 123–183, filler to 186.

On the chimney breast

13. Find the breast's centre: 58.75 cm from its alcove-side corner. Draw it plumb from the floor to 170.

14. Mark the backboard: 55 cm either side of the centre line, from 51.3 to 73.3 high.
15. Mark the two bracket centres 40 cm either side of the centre line.
16. Mark the folded table's top edge at 153.3, and the framed LPs' bottom edges at 165.

Check for cables

17. Run the detector along every line where you'll drill: the cleats, the spacer blocks, the upright brackets, the backboard and the PC shelf.
18. Mark anything it finds in red, and compare with your own photos from step 3. Move a fixing 5 cm rather than drill near a cable.

Check before moving on: Every line level end to end, and the uprights' lines plumb.

Step 6 · Build the base · You · 1 day

MEDIA WALL

Buy first: 4 Base unit (the buy list).

Parts

- Two IKEA BESTÅ frames 60 × 40 × 64
- 18 mm birch ply: bay 3 sides 73 × 40 (2), top and bottom 56.4 × 40 (2); plinth rails 120 × 9 (2), cross-pieces 34.4 × 9 (3)
- Four 4 cm spacer blocks (ply offcuts, 4 × 10 × 1.8)
- Anti-tip brackets: two per bay (6)
- Flat two-compartment trunking, about 2 m
- Plastic shims
- Four connector bolts (furniture joint connectors)

Hardware and fixings

- 4 × 40 and 4 × 50 mm wood screws
- 5 × 60 mm screws and brown wall plugs
- Wood glue

Tools

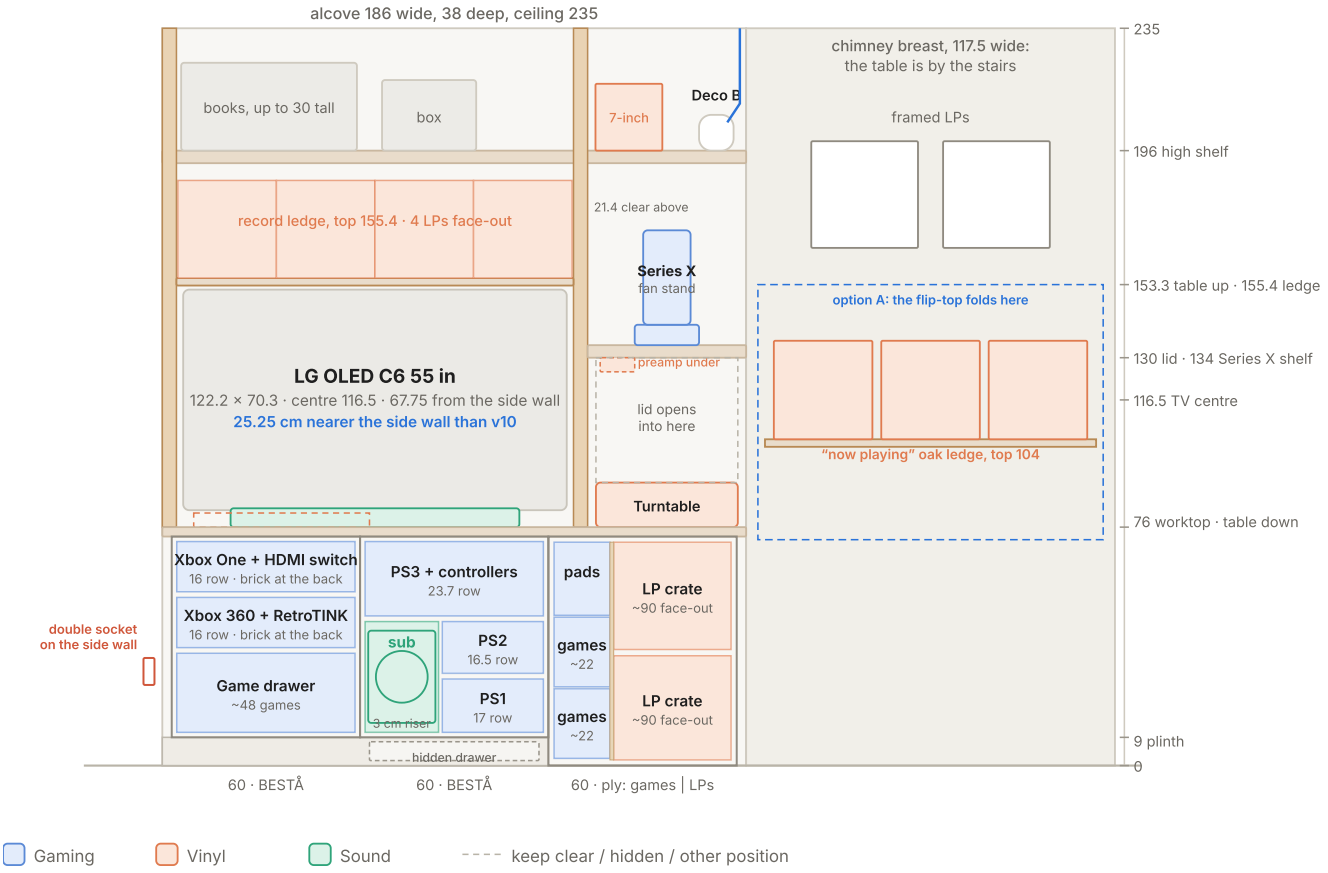
- Drill-driver with pilot and countersink bits
- SDS drill with a 7 mm masonry bit
- Spirit level, 60 and 120 cm
- Combination square
- Clamps
- Tape measure



Every drawer out, in place.

The media wall and the chimney breast from the front, to scale

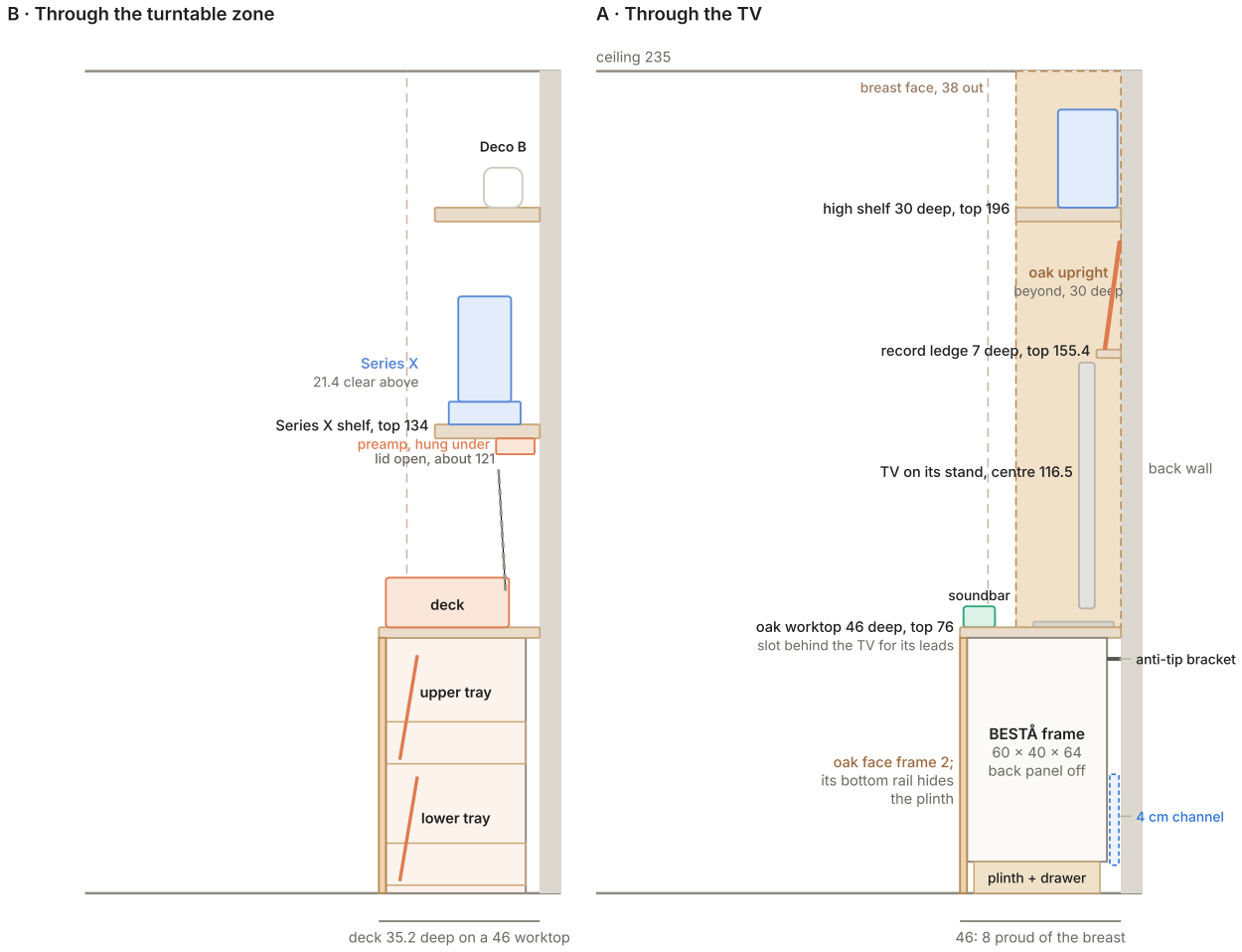
The TV between oak uprights nearer the side wall, the turntable zone on its right, three bays below (bay 3: games beside the LP crates); the “now playing” ledg



Gaming Vinyl Sound keep clear / hidden / other position

How the wall goes together, layer by layer

Side sections to scale, room on the left. Dashed: the upright and the breast's face beyond the cut, and the deck's lid open



Cut list for this step

Part	From	Size (cm)	Qty	Notes
Face frame	Solid oak, hardwax oil	186 × 73 overall, 2 thick, stiles about 5 wide	1 set	End stiles scribed to the walls
Bay 3 sides	Birch ply, 18 mm	73 × 40	2	The record carcass, standing on the floor
Bay 3 top and bottom	Birch ply, 18 mm	56.4 × 40	2	Between the sides; back left open to the channel
Plinth rails	Birch ply, 18 mm	120 × 9	2	Front and back of the ladder frame under bays 1–2
Plinth cross-pieces	Birch ply, 18 mm	34.4 × 9	3	At both ends and under the bay 1–bay 2 joint, so the bay under bay 2 stays clear for the drawer

The plinth under bays 1 and 2

1. Lay the two 120 × 9 rails on edge, 38 cm apart outside, and stand the three 34.4 × 9 cross-pieces between them: one at each end and one in the middle, at 60.
2. Pre-drill, glue and screw through the rails into each cross-piece end with two 4 × 40 screws.
3. Measure both diagonals: when they're equal the frame is square. The plinth is 120 × 38 × 9. The opening between 60 and 120 stays clear: it's the hidden drawer's bay.

4. Stand it in the alcove 3 cm from the side wall and 4 cm off the back wall. Level it with shims, side to side and front to back.

The BESTÅ frames

5. Assemble both frames to IKEA's instructions, but leave the back panels off: the open backs are how the cables and the sub's port breathe.
6. Join the two frames side by side with IKEA's connectors.
7. Stand them on the plinth, their backs flush with its back edge, so their fronts sit 2 cm proud of the plinth (the face frame covers the step).

Bay 3, the carcass for the LP crates and the slim column

8. While the sides are flat, mark the runner lines on their inner faces: on the right side 2.3 and 37.0 cm up from the bottom edge, for the two LP crates; on the left side 2.3, 25.1 and 47.9, for the three slim drawers.
9. Stand the bottom panel between the two sides, flush with their bottom edges, and the top panel flush with their top edges. Pre-drill, countersink, glue and screw with four 4 × 50 screws a joint.
10. Check the diagonals for square. The carcass is 60 × 40 × 73, open at the back.
11. Stand it in its place, 123–183 from the side wall, its top level with the frames' tops at 73. Shim under it if the floor drops away.
12. Clamp it to bay 2 and fit two connector bolts at the front and two at the back.

Fix it to the wall

13. Screw a 4 cm spacer block to the back wall behind the top of each bay, at 68 high, with plugs.
14. Push the whole base back against the blocks.
15. Fit two anti-tip brackets per bay through the blocks into the wall. Do this before anything heavy goes in: a full LP crate pulled out is about 30 kg in front of the unit.
16. Lay a level across all three tops: within 1 mm over 186 cm. Shim and re-check.

The cable channel

17. Lay the flat two-compartment trunking on the floor in the 4 cm channel behind the base: mains on the wall side, data on the other.
18. Run the leads in now: the two mains leads from the side-wall socket enter through the plinth's left end, the Cat6 and Deco B's lead come down the right-hand corner.
19. Keep the channel behind bay 2 empty: the sub's port breathes into it. Switch 2 and the retro 6-way stand on edge behind bay 1.

Check before moving on: The tops level within 1 mm across 186 cm, and nothing rocks.

Step 7 · Fit the worktop · You + a helper · ½ day

MEDIA WALL

Buy first: 5 Oak joinery (the buy list).

Parts

- One solid oak worktop, 300 × 62 × 3, full stave
- A 60 mm brush strip and a 35 mm brush grommet

Hardware and fixings

- 4 × 30 mm screws with washers (9)
- Hardwax oil

Tools

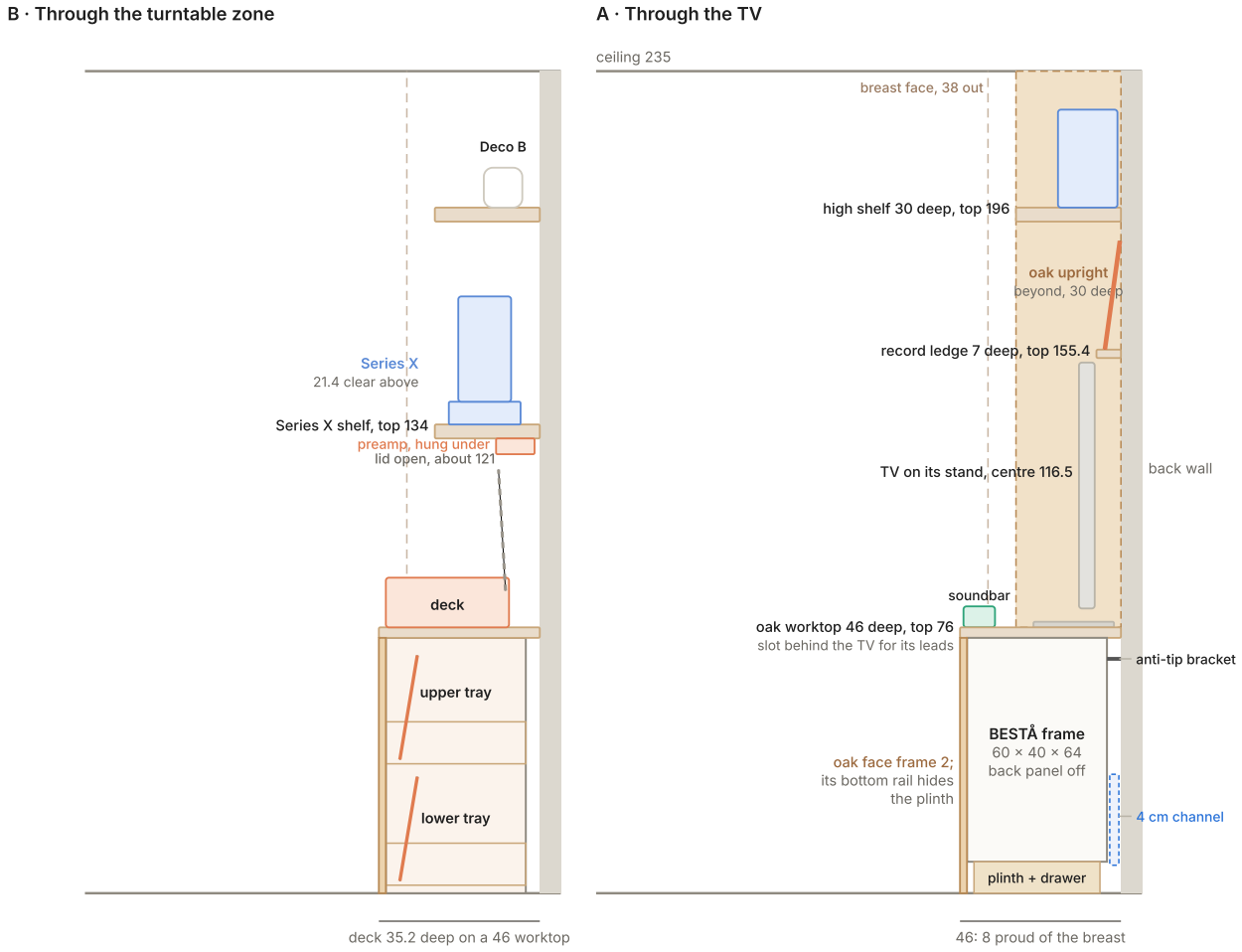
- Hardboard or cardboard, 190 × 50, for the template
- Compass or a pencil on a block, for scribing
- Track saw or circular saw with a guide
- Jigsaw
- Router with a straight bit
- Random-orbit sander, 120 and 180 grit
- Drill-driver
- A helper



The finished media wall.

How the wall goes together, layer by layer

Side sections to scale, room on the left. Dashed: the upright and the breast's face beyond the cut, and the deck's lid open



Cut list for this step

Part	From	Size (cm)	Qty	Notes
Worktop	Solid oak, hardwax oil	186 × 46 × 3	1	From one 300 × 62 full-stave worktop; chamfered front

Make a template

1. Cut the hardboard to about 190 × 50 and push it into the alcove at worktop height, its back edge against the back wall.
2. Set the compass to the widest gap at one end and run it down the side wall, so the pencil copies the wall's line onto the board. Cut to the line. Do the other end the same way.
3. Mark the front edge at 46 cm from the back wall all along. The template is now the worktop's exact shape.

Cut the oak

4. Lay the board best face up, grain running along its length. Lay the template on it and draw round it.
5. Cut just outside the line with the track saw, then sand or plane to the line.
6. From the offcut, cut the dining wall's backboard: 110 × 22 × 10.3.
7. Chamfer the worktop's top front edge with a 3 mm chamfer, sanded by hand.

Cut the openings

8. The TV slot: 3 cm wide, from 1 to 4 cm in from the back edge, running 27.75–107.75 from the side wall. Clamp a batten as a guide and rout it in passes, or drill each end and jigsaw between. Push the brush strip in.
9. The grommet hole behind the deck: 35 mm, its centre 2.5 cm from the back edge and 160.75 from the side wall.

Finish and fix

10. Sand to 120, then 180. Oil every face: top, underside, both ends and the cut edges of the slot. Two coats, a night between them.
11. Two of you lift it in and push it back to the wall. It should touch both side walls along their length.
12. Inside each bay, drill three 5 mm holes up through the top panel, elongated front to back to about 12 mm long, so the oak can move with the seasons.
13. Screw up through them into the worktop with 4 × 30 screws and washers. They must not come out through the top: check the length against a scrap.

Check before moving on: Tight to both walls, and the slot sits over the channel.

Step 8 · Fit out the base · You · 1–2 days

MEDIA WALL

Parts

- Four BESTÅ shelves 56 × 36
- Game drawer: 12 mm ply sides 35 × 15.5 (2), back 51.9 × 15.5, front panel 51.9 × 6; 9 mm base 52.5 × 34.4; 6 mm divider 51.9 × 9, bin walls 12.6 × 9 (2), bin floors 16 × 12.6 (2); oak lip 56.8 × 6.5 × 2; follower blocks
- LP crates (×2): 12 mm base 34.9 × 40; 9 mm sides 40 × 8 and ends 33.1 × 8; oak lip 36.3 × 5 × 1.5; felt
- Slim drawers (×3): 12 mm base 15.3 × 40; 6 mm sides 40 × 10, back and front 14.1 × 10; oak lip 16.5 × 5 × 1.5; a 2 m USB-C lead and a grommet for the top one
- Bay 3's divider: 12 mm ply 69.4 × 38
- Plinth drawer: 12 mm sides 34 × 4.4 and ends 49.6 × 4.4, 6 mm base 52 × 34, oak front 54 × 6 × 2
- Bay 2: 12 mm divider 35.3 × 38, part shelf 32.2 × 36, the sub's isolation riser about 22 × 28 × 3
- The oak face frame, 2 cm thick: stiles about 5 wide, rails; 186 × 73 overall

Hardware and fixings

- Runners: two pairs of 400 mm (68–70 kg) for the LP crates, three pairs of 400 mm light-duty for the slim drawers, one pair of 350 mm soft-close, one pair of 300 mm
- 3.5 × 16 mm screws for the runners (every hole)
- Four shelf pins for the part shelf
- Two small steel angle brackets for the divider
- Panel pins, wood glue, felt

Tools

- Drill-driver
- Router with a 6 mm straight bit (the game drawer's base groove)
- Combination square
- Spirit level
- Clamps
- Mitre saw or tenon saw

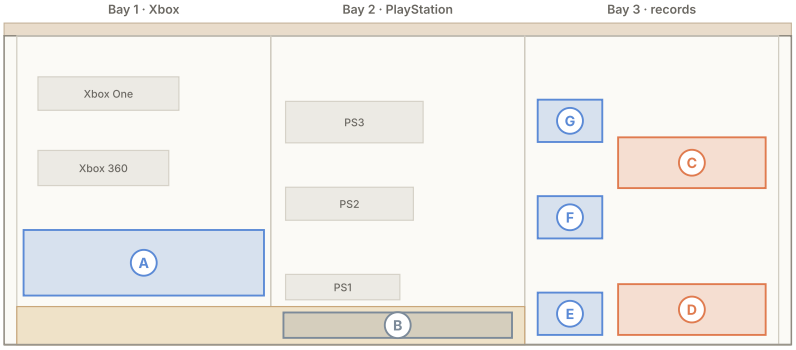


Every drawer out, in place.

The drawer layouts: where each drawer is and what goes in it

Seven drawers in the alcove base, all to scale: A the game drawer, B the hidden plinth drawer, C and D the LP crates, E to G the slim drawers beside them

Where each drawer is: the alcove base from the front



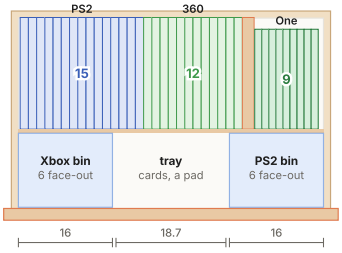
186 wide: 3 + 60 + 60 + 60 + 3 · worktop at 76 · front view, 3.6 px per cm

- **A · Game drawer: bay 1, bottom row (11.6–27.1 up)**
- **B · Plinth drawer: hidden under bay 2 (1.6–7.6 up)**
- **C · Upper LP crate: bay 3, right (37–69.1 up)**
- **D · Lower LP crate: bay 3, right (2.3–34.4 up)**
- **E · Slim drawer, games: bay 3, left (2.3 up)**
- **F · Slim drawer, games: bay 3, left (25.1 up)**
- **G · Slim drawer, pads and remotes: bay 3, left (47.9 up)**

Open one LP crate at a time: full, one is about 30 kg.
Anti-tip brackets fix the base to the wall first.
Every front: a plain oak lip, finger pull underneath.

A · Game drawer

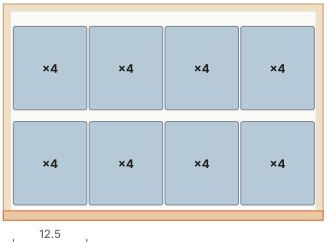
box 54.3 × 35 · 350 mm soft-close runners · rolls out 30



back: 36 spine-up, titles on top (19 cm cases on edge)
front: 12 face-out · about 48 games in all

B · Hidden plinth drawer

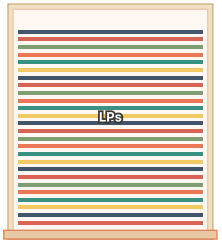
box 52 × 34 × 5 inside · 300 mm runners · finger pull



32 PS1 games flat: 8 stacks of 4 jewel cases
or remotes, spare pads, cables and manuals

C and D · LP crates

box 34.9 × 40 · 400 mm runners



trough 33.1: snug, no dead space

face-out, leaning back 10°
~90 LPs each · two crates

E, F, G · Slim drawers

15.3 × 40 · 400 mm



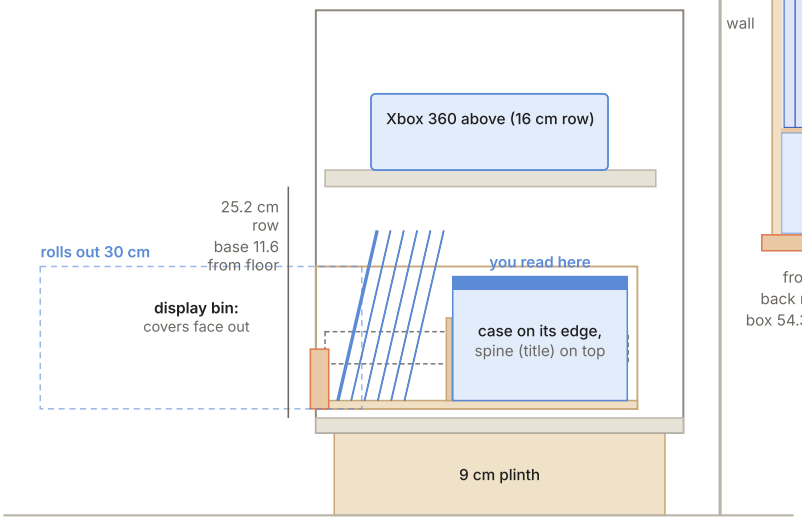
E, F: ~22 games
each, face-out
G: 4 pads, remotes,
a charging lead

Plans drawn from above at 5 px per cm, front (the room) at the bottom; the elevation at 3.6 px per cm. Sizes: BESTÅ frame 60 × 40 × 64 (IKEA);
game cases 19 × 13.5 × 1.4 and 17 × 13.5 × 1.2; jewel case 14.2 × 12.5 × 1; LP sleeve 31.4; pads 16 × 10.6 × 6.6; runners Emuca 400 mm, Accuride 350 mm (see the fit proof).

The game drawer: covers on show at the front, every title readable from above at the back

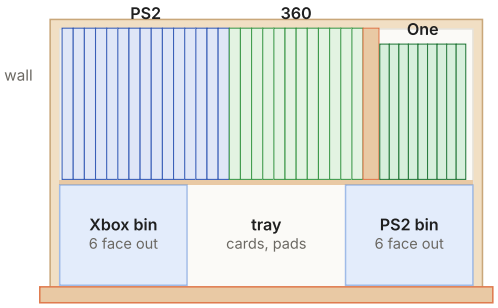
Bay 1, bottom row. Section left, plan right. Nothing stands behind anything else

Side section through bay 1



box 35 cm deep on 350 mm full-extension soft-close runners
5 cm free behind the box, then the 4 cm cable channel
bins: 19 cm cases leaning 13°, about 18.5 cm tall

Top view, drawer pulled out

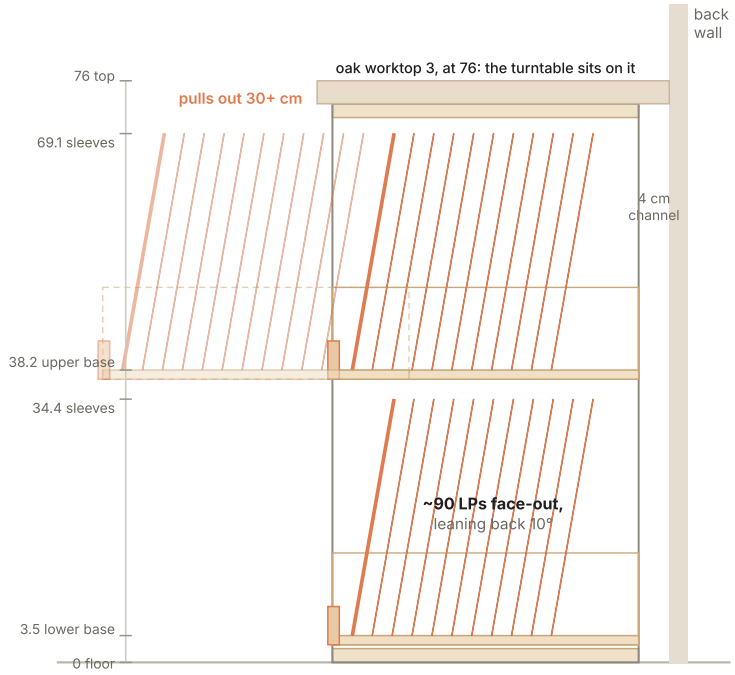


front: oak lip with finger pull, matching the record drawer
back row: 15 PS2 + 12 Xbox 360 + follower block + 9 Xbox One
box 54.3 × 35 cm outside · 51.9 cm inside · 16 cm bins, 18.7 cm tray

Bay 3: two LP crates under the turntable, a slim game column beside them

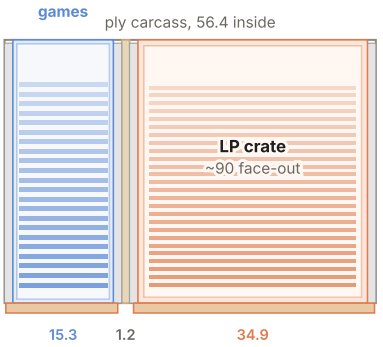
Section left through the LP crates, plan right. Pull a crate and flip through the covers like a record shop browser; the slim drawers flip the same way for games

Side section through bay 3



carcass 18 mm birch ply, 60 × 40 × 73, back open to the channel · crates 40 deep on 400 mm runners
12 mm base, 9 mm sides, 5 cm oak lip · 2.6 and 2.1 cm finger room (top panel at 71.2)

Bay 3 from above (v16)



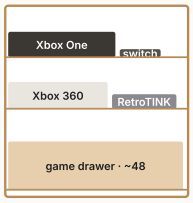
LP crate 34.9 × 40, a 33.1 trough: LPs snug, no dead space
slim drawers 15.3 × 40, 14.1 inside: DVD-size cases face-out
12 mm ply divider · 12.7 mm runners each side
an LP crate is about 30 kg full: open one at a time

Storage options: what fits each space, to scale

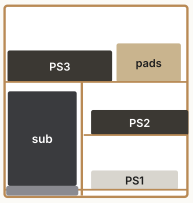
Front views and views from above of every storage space in the wall, with counts, and where the overflow would go

Bays 1 and 2: inside each BESTÅ frame

56.8 wide × 40 deep inside; rows as planned, front view



Bay 1 · Xbox



Bay 2 · PlayStation + sub

Swap any 16 cm row for two KUGGIS boxes, 26 × 35 × 15, side by side (52 of 56.8 wide).

Bay 3 from above: LP crate and slim drawer

56.4 inside: a 15.3 slim drawer, the divider, a 34.9 LP crate



games | divider | LPs: no dead space

≈ 90 LPs a crate, two crates

= 180 LPs, all face-out

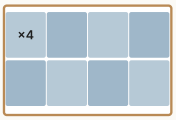
beside them 44 games face-out

and a drawer for pads and remotes

trough 33.1: thick outer sleeves fit

Hidden plinth drawer, from above

box 52 × 34 × 5 inside, under bay 2; finger pull, not push-to-open



32 PS1 games flat

jewel cases 14.2 × 12.5 × 1, 8 stacks of 4

or: remotes, spare pads, cables, manuals

Series X shelf, front view

50.5 wide, 134 to the high shelf at 192: 58 clear



15 air each side, 21.4 above

Microsoft: at least 4 above, 10–15 recommended all round

Under it: the deck, lid opens to 121

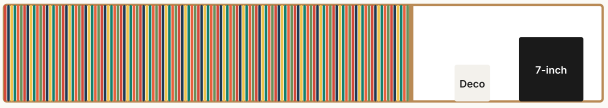
High shelf, front view: as planned, or as LP overflow

186 long, 30 deep, 39 clear to the ceiling; the right-hand 50.5 keeps Deco B and the framed 7-inch



about 1 m of books, a lidded box, a print

As planned



≈ 380 LPs spine-out at about 3.3 mm each; a 31.4 sleeve overhangs the 30 cm shelf by 1.4

As LP overflow

PC shelf, from above

82 along the wall × 28 out, at 155.4; keyboard tray in a 7 cm housing under it



side wall along the top · alcove to the right

Keyboard tray: 300 mm runners, rolls 25 cm out; the housing stops 12 short of the sofa end so the cables drop past it. Full-size wireless keyboard and mouse.

If the collection outgrows the wall

two overflow options, both face-out or spine-out to match

1 · LPs spine-out on the high shelf: about 380 more

No building; books move to the second option's top.

2 · A second record unit, west of the sofa, under the window

Another LP column like bay 3 (40 × 46 × 73, two crates): 180 LPs.

At x 300–360 it leaves 123 cm to the table set for six (100 to walk).



front view, 60 × 73 (shown at half height)

All to one scale, 2.2 px per cm. Sizes: KUGGIS box 26 × 35 × 15 (IKEA); jewel case 142 × 125 × 10 mm (Wikipedia: Optical disc packaging); consoles, sleeves, Series X and the trays from the fit proof (section 3 of the plan).

Cut list for this step

Part	From	Size (cm)	Qty	Notes
Drawer fronts	Solid oak, hardwax oil	game drawer 56.8 × 6.5 × 2 · plinth drawer 54 × 6 × 2	1 · 1	Finger pull cut underneath
LP crate lips · slim drawer lips	Solid oak, hardwax oil	36.3 × 5 × 1.5 · 16.5 × 5 × 1.5	2 · 3	Cut to the face frame's openings, less 3 mm (v16)

Follower blocks · Xbox One lift block	Solid oak, hardwax oil	19 × 12 × 2 · 15 × 12 × 2	2 · 1	Offcuts, felted underneath
Game drawer sides	Birch ply, 12 mm	35 × 15.5	2	
Game drawer back · front panel	Birch ply, 12 mm	51.9 × 15.5 · 51.9 × 6	1 · 1	
LP crate bases · slim drawer bases	Birch ply, 12 mm	34.9 × 40 · 15.3 × 40	2 · 3	The sides and ends stand on them (v16)
Bay 3 divider	Birch ply, 12 mm	69.4 × 38	1	17.8 in from the left side, the full height inside (v16)
Plinth drawer sides · ends	Birch ply, 12 mm	34 × 4.4 · 49.6 × 4.4	2 · 2	A 52 × 34 × 5 box
Bay 2 divider · part shelf	Birch ply, 12 mm	35.3 × 38 · 32.2 × 36	1 · 1	The sub's column on the left; the PS2 over the PS1 on the right
Game drawer base	Birch ply, 9 mm	52.5 × 34.4	1	In a 6 mm groove
LP crate sides · ends	Birch ply, 9 mm	40 × 8 · 33.1 × 8	4 · 4	Two of each per crate: a 33 cm trough (v16)
Game drawer cross divider	Birch ply, 6 mm	51.9 × 9	1	13 cm back from the front
Game drawer bin walls · bin floors	Birch ply, 6 mm	12.6 × 9 · 16 × 12.6	2 · 2	Floors raked so the cases lean about 13°
Plinth drawer base	Birch ply, 6 mm	52 × 34	1	Screwed under the box
Slim drawer sides · backs and fronts	Birch ply, 6 mm	40 × 10 · 14.1 × 10	6 · 6	Three boxes 15.3 wide, 14.1 inside (v16)
LP crate runners · slim drawer runners	Hardware	400 mm full extension, 68–70 kg a pair · 400 mm light duty	2 pairs · 3 pairs	Into bay 3's sides and the divider (v16)
Game drawer runners	Hardware	350 mm full extension, soft-close	1 pair	Accuride DZ3832EC or Screwfix 964VN

Plinth drawer and keyboard tray runners	Hardware	300 mm full extension · 300 mm light duty	1 pair · 1 pair	
Sub isolation riser	Hardware	dense foam or sorbothane, about 22 × 28 × 3	1	Under the sub in bay 2

The console shelves

1. Bay 1, from the top: rows of 16 (Xbox One), 16 (Xbox 360), then the game drawer's 25.2. Put the shelves on the pin holes nearest those heights.
2. Bay 2: the PS3's row of 23.7 at the top, on a full-width shelf. Below it, the PS2 and the PS1 sit beside the sub (next part).

Bay 2's sub column

3. Stand the 12 mm divider on the bay's floor, 23.4 cm from the bay's left inner face, tight under the PS3 shelf. Check it's plumb.
4. Fix it with one small angle bracket at the bottom (into the frame's bottom panel) and one at the top (into the PS3 shelf's underside).
5. Drill two 5 mm pin holes in the divider's right face at the height of the frame's pin holes for the PS2/PS1 shelf. Fit the part shelf on four pins.
6. Pair the sub with the soundbar now: its PAIR button is on the back, and it's hard to reach once it's in.
7. Lay the isolation riser on the column's floor, its front edge flush with the frame's front.
8. Stand the sub on it with the grille facing the room, centred: 1 cm to the divider, 1 cm to the frame's side. Slide it forward until its front is flush with the face frame's front (2 cm proud of the frame). Run its lead out of the back.
9. Check it touches nothing but the riser. If it does, a sheet of felt between, or nudge it.

The game drawer (bay 1, bottom row)

10. Fit the 350 mm runners to the frame's sides, level, so the drawer's base will sit about 1 cm above the frame's bottom panel. Drill pilot holes: BESTÅ sides are particleboard. If a screw won't hold, glue a 12 mm ply packer behind the runner.
11. Rout a 6 mm deep groove 1 cm up from the bottom on the inner faces of the sides, back and front panel, 9 mm wide, for the base.
12. Glue and screw the box together round the base: outside 54.3 × 35. Check the width against the runner maker's figure (usually the opening less 25.4 mm) before the glue sets.
13. Fit the cross divider 13 cm back from the front, and the two bin walls 16 cm in from each side. Glue in the bin floors with a slight back-rake, so the cases lean about 13°.
14. Felt the base and make the follower blocks.
15. Fit the runners' inner members to the box, hang it, and check it runs true empty.

Bay 3: the divider, the LP crates and the slim drawers (v16)

16. Stand the 12 mm divider in the bay, its left face 17.8 cm from the left side's inner face, the full height inside. Check it's plumb and parallel to the side at front and back (measure both), then fix it with two small angle brackets top and bottom.
17. Mark the runner lines on the divider's faces: on its right face 2.3 and 37.0 (the crates), on its left face 2.3, 25.1 and 47.9 (the slim drawers), matching the lines on the sides.
18. Screw the crates' 400 mm runners to the right side and the divider's right face; the slim drawers' to the left side and the divider's left face. Use every hole: a full crate is about 30 kg.
19. For each LP crate, glue and pin the two 40 × 8 sides and two 33.1 × 8 ends on top of the 34.9 × 40 base. The trough is 33 inside: snug for 31.4 sleeves, like a record shop's crate.
20. Glue a raked rest (an oak offcut) at the back of each crate, so the sleeves lean back about 10°. Felt the trough.

21. For each slim drawer, glue and pin the 6 mm sides, back and front onto the 15.3 × 40 base: 14.1 inside, for 13.5 cm game cases face-out. In the top one, glue a 6 mm strip down the middle (two pads on edge each side) and drill a 20 mm hole in the back for the USB-C lead's grommet.
22. Hang each box and run it in and out empty, then with ten records or ten games. Feed the charging lead from a USB plug on the 10-gang down the channel, with a loop of slack so the drawer opens fully.

The hidden plinth drawer

23. Make the box: the two 34 × 4.4 sides, the two 49.6 × 4.4 ends between them, the 6 mm base screwed underneath. It's 52 × 34 × 5.
24. Pack the plinth's inner faces under bay 2 with ply strips so the 300 mm runners sit the box's width plus their allowance apart, and fit the runners with the box 1.6 cm off the floor.
25. Felt the runners: the drawer sits under the sub.

The face frame and the drawer fronts

26. Cut the stiles (about 5 wide, and one 2 wide over bay 3's divider) and the rails from 2 cm oak. The bottom rail covers the plinth, 9 tall under bays 1–2, with an opening 54 × 6 for the plinth drawer, 1.6 up, under bay 2.
27. Scribe the two end stiles to the walls, then glue and pin the frame to the frames' fronts and bay 3. Notch the bottom rail's left end for the two socket leads.
28. Screw the oak lips onto the game drawer, the two LP crates, the three slim drawers and the plinth drawer from inside each box, with a 2–3 mm gap all round to the face frame.
29. Cut each finger pull underneath its lip: a cove routed along the back of the bottom edge.

Check before moving on: Every drawer runs smoothly, the two crates clear each other by 2 cm, and the slim drawers by 2.8.

Step 9 · Uprights and shelves · You + a helper · 2–3 days

MEDIA WALL

Parts

- Two oak uprights, 165 × 30 × 4.5 (cut to fit)
- High shelf: 128.9 × 30 × 4 and 51.7 × 30 × 4
- Record ledge 128.9 × 7 × 2.4 and its lip
- Series X shelf 55 × 30 × 4
- Two oak cleats for the reveal, 30 long
- The “now playing” ledge 105.5 × 7 × 2.4 and its lip, two concealed brackets
- Oak wedges (2 + spares), 10 mm oak dowel for pegs and the uprights' dowels

Hardware and fixings

- Four concealed upright brackets
- 5 × 60 mm screws and plugs
- Wood glue
- LED strip in its aluminium channel (from the lighting phase)

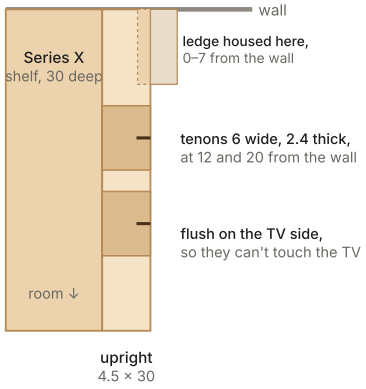
Tools

- Router with a straight bit and a guide batten
- Chisels 6, 12 and 25 mm, and a mallet
- Japanese pull saw
- Drill with a 10 mm brad-point bit and a 20 mm bit
- Dowel jig or dowel centres
- Combination square, marking knife
- Spirit level and laser
- Clamps
- A helper

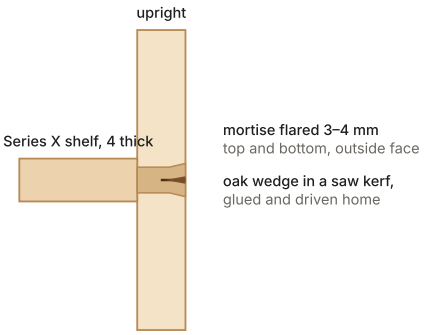
Artisan oak: how the joints work

Drawn to scale at 7 px per cm. The joints carry the load and are left on show, finished in hardwax oil

A · Wedged through-tenons, from above

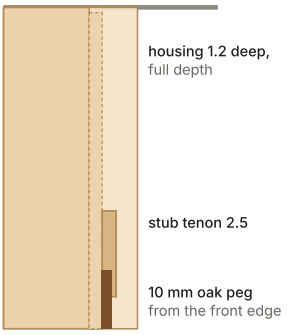


B · The same joint from the front



The kerf runs flat, so the wedge spreads the tenon up and down, along the upright's grain, and can't split it.

C · Housed shelf, pegged

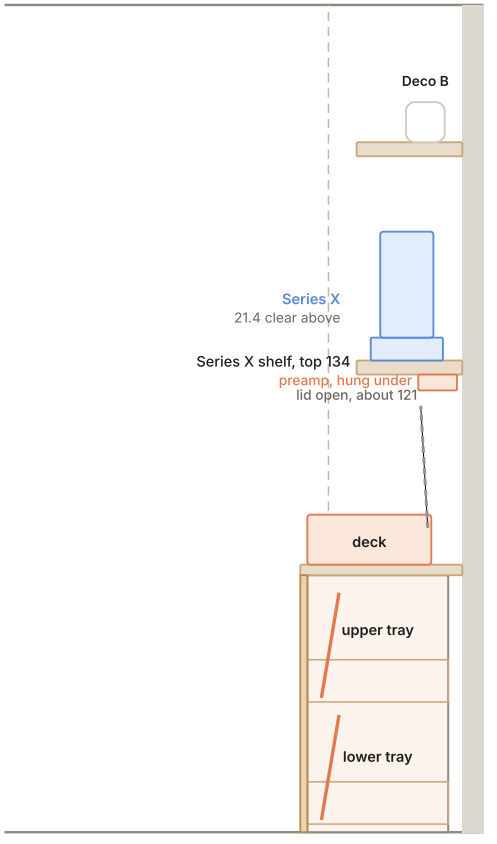


- Through-tenons (A, B): the Series X shelf (top 134) passes through the right upright on two tenons, 12 and 20 cm from the wall, wedged and planed flush: proud, they'd touch the TV
- Housed and pegged (C): the three lengths of the high shelf (top 196) and the record ledge (top 155.4). The peg shows as a dark dot on the upright's face.
- Uprights: 159 long, worktop to ceiling, each on two 10 mm dowels into the worktop, scribed to the ceiling, with two concealed brackets. The left one also screws to the side wall, and

How the wall goes together, layer by layer

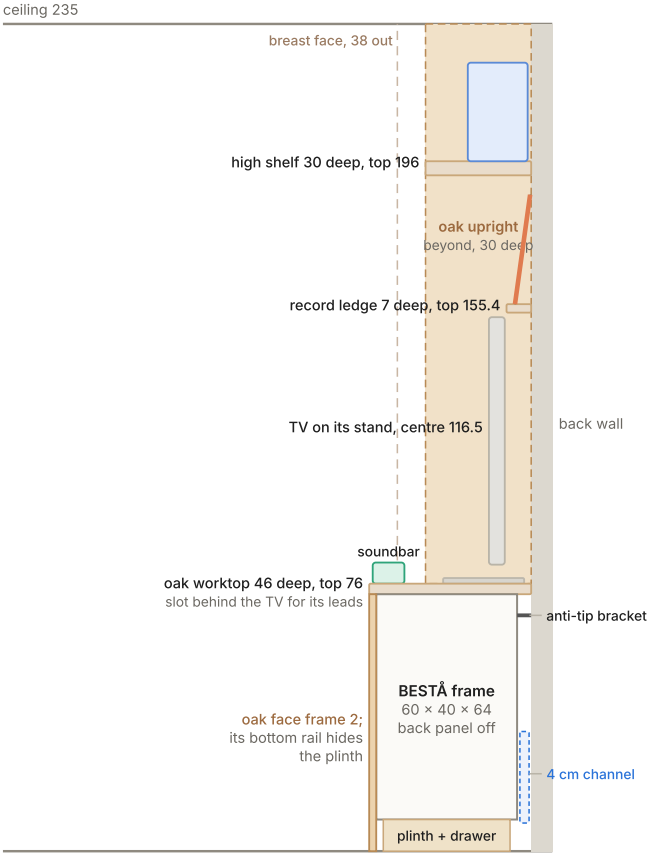
Side sections to scale, room on the left. Dashed: the upright and the breast's face beyond the cut, and the deck's lid open

B · Through the turntable zone



deck 35.2 deep on a 46 worktop

A · Through the TV



46: 8 proud of the breast

Cut list for this step

Part	From	Size (cm)	Qty	Notes
Uprights	Solid oak, hardwax oil	159 × 30 × 4.5	2	Buy 165 long and scribe to the ceiling; housings, mortises, LED and cable grooves cut on the bench
High shelf, long length	Solid oak, hardwax oil	128.9 × 30 × 4	1	Housed 1.2 into both uprights and pegged
High shelf, short length	Solid oak, hardwax oil	51.7 × 30 × 4	1	Housed into the right upright; cleat on the reveal
Series X shelf	Solid oak, hardwax oil	55 × 30 × 4	1	Two through-tenons 6 × 2.4, wedged flush
Record ledge	Solid oak, hardwax oil	128.9 × 7 × 2.4	1	Housed and pegged
Ledge lips	Solid oak, hardwax oil	128.9 × 1.5 × 0.9 · 126 × 1.5 × 0.9	1 · 1	Glued along the ledge's and the run's front edges
“Now playing” ledge	Solid oak, hardwax oil	105.5 × 7 × 2.4, lip 1.5	1	On the breast at 104, two concealed brackets (v16; with the table by the stairs)
Wedges · pegs · dowels	Solid oak, hardwax oil	6 × 3, tapering · 10 mm dowel, 5 long · 10 mm dowel	2 + spares · 6 + 6 · 8	Glued

Cut the uprights to length

1. Measure from the worktop to the ceiling at each upright's position. The plan says 159; ceilings vary.
2. Cut each upright 2 mm under its measurement, then scribe its top to the ceiling as you did the worktop.
3. Mark each upright's faces: which side faces the TV, and the back edge that goes against the wall.

The housings (both uprights)

4. Measure from each upright's foot (the end on the worktop). The high shelf's housing is 116–120 up, the full 30 depth, 1.2 deep, on the TV-side face of both uprights.
5. The record ledge's housing is 77–79.4 up, 0–7 from the back edge, 1.2 deep, on the TV-side face of both.
6. On the right upright's other face (the turntable side), a second high shelf housing at 116–120, full depth, for the short length.
7. Clamp a straight batten as a guide, set the router 4 mm deep and rout, then step down to 12 mm in passes. Square the ends with a chisel. Practise on an offcut first.

The through-mortises (right upright)

8. Two mortises, each 6 wide and 2.4 tall, at 54.8–57.2 up from the foot, 12–18 and 20–26 cm from the back edge.
9. Mark them on both faces with a square and a knife. Drill most of each out with the 20 mm bit, halfway from each face, then chisel to the lines.
10. On the TV-side face, flare the top and bottom of each mortise outwards 3–4 mm, for the wedges.

The tenons (Series X shelf)

11. On the shelf's west end (the end that goes through the upright), mark two tenons 6 wide and 2.4 thick, centred in its 4 cm thickness, 12–18 and 20–26 from its back edge, 4.5 long.
12. Saw the shoulders and cheeks, and cut away the rest of the end back to the shoulder line.
13. Saw a kerf 3 cm deep across the middle of each tenon, parallel to the shelf's faces.
14. Test-fit through the mortises: snug, pushed in by hand.

The grooves

15. On each upright's TV-side face, rout a groove for the LED channel (usually about 1.7 wide × 0.8 deep: check yours), 2 cm in from the front edge, from 2 cm above the worktop to just under the high shelf's housing.
16. On the right upright's back edge, rout a 2 × 1.5 cable groove the full length, for Deco B's 12 V and patch leads, the Series X's HDMI and power, and the vinyl lead.
17. Notch both uprights' tops for the ceiling LED strip.

Dry-fit, then fix

18. Lay each upright on the floor with its shelves in their housings. Check every joint closes before any glue.
19. Drill two 10 mm holes 3 cm deep in each upright's foot, and matching holes in the worktop at the marked positions (dowel centres make this exact). Glue the dowels into the uprights only, so they can come out.
20. Lay Deco B's leads and the other cables into the right upright's back groove.
21. Stand the left upright on its dowels against the side wall. Screw it to the side wall through three countersunk holes, with plugs, and plug the holes with oak.
22. Stand the right upright on its dowels, check it's plumb with the laser, and fix it to the back wall with the concealed brackets.
23. Slide the long high shelf and the record ledge into their housings from the front.
24. Screw a 30 cm cleat to the breast's side face (the reveal) for the short high shelf, and another at the Series X shelf's underside height. Slide the short high shelf in from the front onto its cleat and into its housing.
25. Feed the Series X shelf's tenons through the mortises from the turntable side, onto its cleat. Glue the wedges and drive them into the kerfs from the TV side. When dry, saw them flush and plane: proud, they'd touch the TV.
26. Pegs: drill 10 mm through each upright's face into the housed shelf ends, 4 cm deep, about 4 cm from the front edge. Glue in a peg and saw it flush.

The “now playing” ledge on the breast (v16)

27. Cut it 105.5 × 7 × 2.4 from the shelf board's offcut and glue a 1.5 cm lip along its front edge, so a sleeve stands against the wall without sliding.
28. Mark a level line on the chimney breast at 101.6 (the ledge's underside), 6 cm in from each side of the breast.
29. Fit its two concealed brackets to the maker's instructions, slide the ledge on, and stand three LPs on it.

Check before moving on: Uprights plumb, shelves level, the TV slides in with 2 cm each side, and the “now playing” ledge is level with an LP standing on it.

Step 10 · The side-wall run and the PC end · You · 1 day

MEDIA WALL

Parts

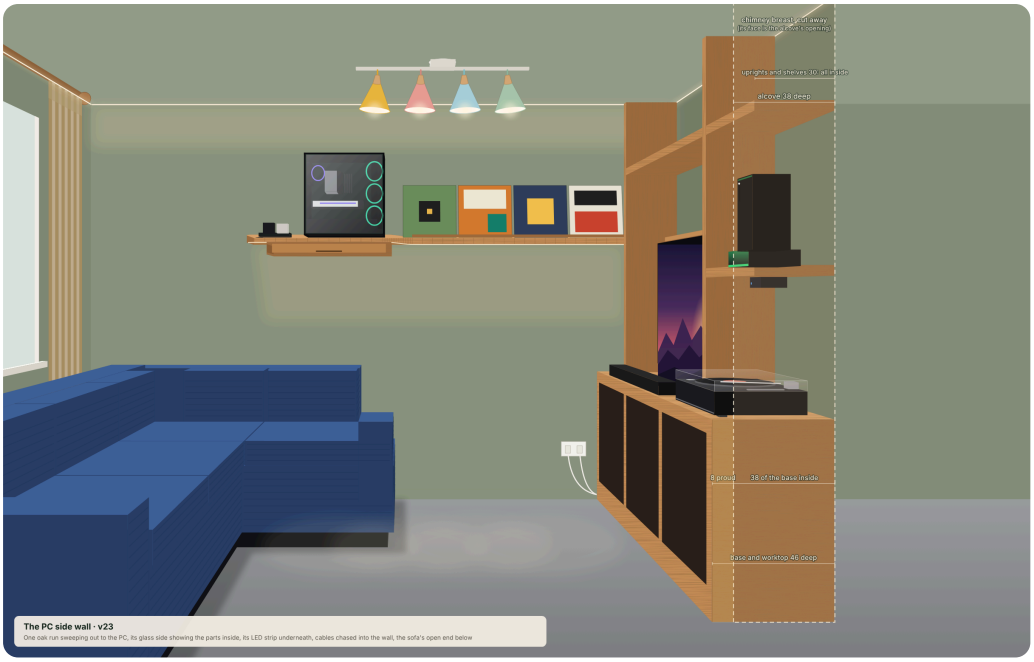
- Wrap ledge 134 × 7 × 2.4 and its lip
- PC shelf 82 × 28 × 4
- Two oak cleats, 2 cm
- Keyboard tray: oak housing 70 × 28 × 7 (sides 7 × 2), 12 mm ply tray 54 × 26 with an oak lip

Hardware and fixings

- Two concealed shelf brackets rated 40 kg+ a pair
- 300 mm light-duty runners, one pair
- An oak-coloured brush grommet
- 10 mm dowels (2)
- 5 × 60 mm screws and plugs

Tools

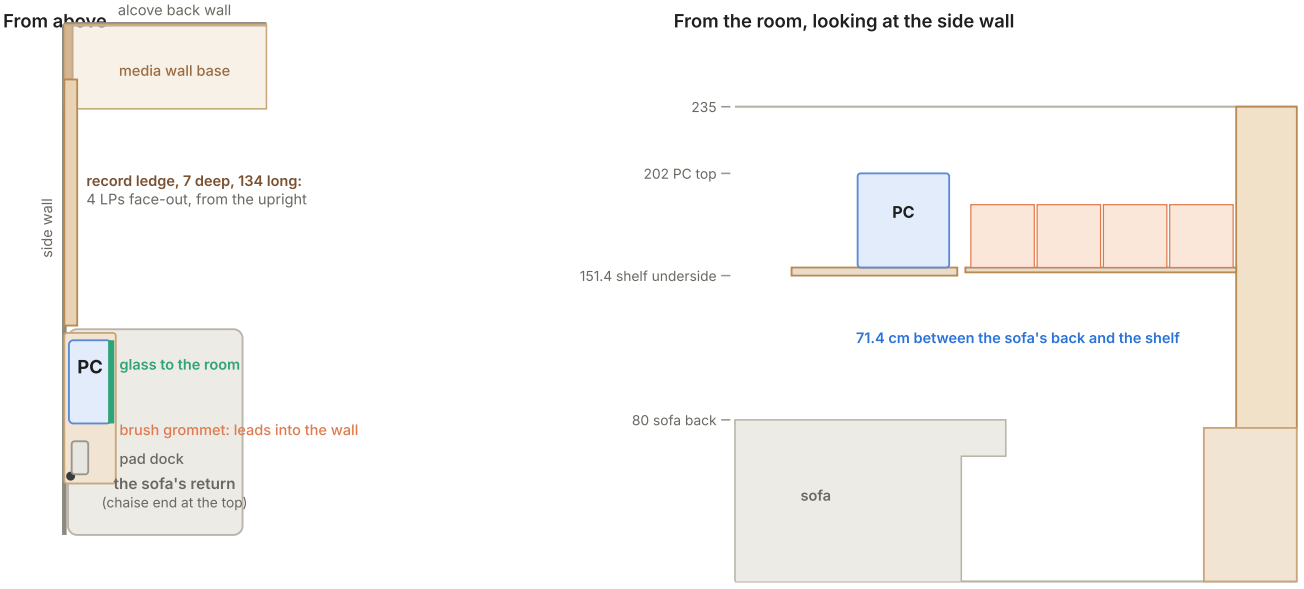
- SDS drill
- Drill-driver
- Dowel jig
- Spirit level and laser
- Jigsaw
- Clamps



The side-wall run and the PC end.

The side wall: the ledge wraps the corner, the PC sits above the sofa

One oak line at 155.4 cm: the record ledge from the upright, then the PC shelf over the end of the sofa, clear of the TV's sightlines



Cut list for this step

Part	From	Size (cm)	Qty	Notes
Side-wall run (v15)	Solid oak, hardwax oil	220 long × 7 sweeping to 28 × 4, from one 225 × 30 × 4 board	1	On an oak cleat, concealed brackets under the PC end, dowelled to the left upright; LED groove under the front edge
Keyboard tray housing	Solid oak, hardwax oil	70 × 28 × 7, sides 7 × 2	1	Screwed under the PC shelf
Keyboard tray	Birch ply, 12 mm	54 × 26	1	Oak lip on its front edge

Concealed brackets	Hardware	for the uprights; two for the PC shelf, 40 kg+ a pair	1 set	

The cleats

1. Screw the first cleat to the side wall with its top on the 153 line, from the left upright's front edge (30 cm out from the back wall) to 164 cm out. Plugs every 25 cm.
2. Screw the second with its top at 151.4 (the PC shelf's underside), from 168 to 250 cm out.

The wrap ledge

3. Drill two 10 mm dowel holes in the ledge's end and matching holes in the left upright's front edge, using the jig.
4. Glue the dowels, push the ledge onto them and down onto its cleat, and screw up through the cleat into it.
5. Glue the lip along its front edge.

The PC shelf and its tray

6. Fit the concealed brackets into the wall at the shelf's height, through the cleat, to the maker's instructions.
7. Cut the grommet hole in the shelf's back corner, over the top of the chased conduit.
8. Screw the keyboard tray's housing to the shelf's underside, stopping 12 cm short of the sofa end so the cables drop past it. Fit the runners inside it and slide the tray in.
9. Slide the shelf onto its brackets and down onto the cleat. Lean on it: it shouldn't move.

The PC's leads

10. Fit the brush grommet. Pull the HDMI (active optical: the end marked "source" at the PC) and the Cat6 down the 38 mm conduit with its draw wire.
11. Take the plug off the C13 lead, feed its bare end down the 25 mm conduit, and at the bottom fit the rewireable plug: brown to L, blue to N, green-and-yellow to E, the cord grip on the outer sheath, a 13 A fuse.
12. At the bottom, all three run behind the skirting into the plinth's notch and the channel; the mains lead goes on to the 10-gang behind the TV.

The PC's extras (v17)

13. Screw the powered USB hub under the side run's front edge, below the PC, ports facing the sofa. Its adapter plugs into the 10-gang along the PC's mains route; its lead runs up through the grommet to the PC.
14. Remote power button: with the PC unplugged, push the receiver card into a spare slot and plug its lead onto the motherboard's power-switch pins (the case's own button lead plugs into the card's pass-through). Stick the push button to the wrap ledge's underside, by the sofa end.
15. Lay the magnetic dust filter on the PC's top exhaust; brush it off once a month.

Check before moving on: One level line with the ledge over the TV, the sweep runs smooth with no flat spots, and the PC end holds your weight leaning on it.

Step 11 · The flip-top dining table · You + a helper · 1–2 days

DINING TABLE

Buy first: 6 Dining wall (the buy list).

Parts

- Solid oak: two panels, each a centre 101 × 80 × 2.7 (edge-glued staves) with two breadboard ends 80 × 7 × 2.7: the top and the half; the backboard 110 × 22 × 3 and the hinge rail 110 × 4 × 7.3 (from step 7), two turn buttons 6 × 2.5 × 2.2
- The gate leg: two tapered oak legs 73.3 × 4.5 to 3 and an apron 94 × 7 × 2

Hardware and fixings

- Two Häfele Hebgo 287.43.428 folding brackets
- Union PowerLoad 100 × 88 lift-off hinges, one 3-pack, all the same hand
- Three Lee Valley 40574 narrow flip-top hinges (1.27 × 7.0) and their screws
- Two 75 mm steel butt hinges and two folding locking stays for the gate
- A magnetic catch
- 5 × 70 mm screws and brown plugs
- Felt pads
- Hardwax oil, glue, 10 mm oak dowel for pegs

Tools

- Router with a straight bit and a guide
- Drill with a 10 mm brad-point bit
- Jigsaw and sander
- Chisels
- SDS drill
- Spirit level
- Clamps
- A helper

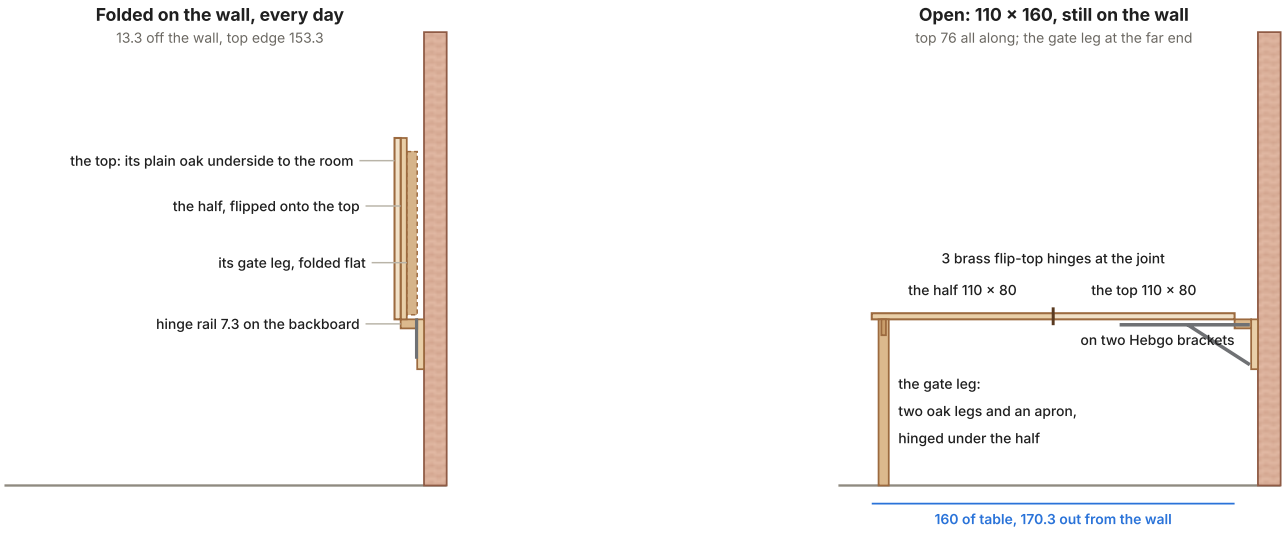


The flip-top open under the shelf by the stairs.

The flip-top table: folded, and open

In section through its centre line, to scale, the wall on the right of each. It never comes off the wall, and there's no leaf to store.

1. To open: turn the two buttons and lower the pair onto the brackets as they swing out and lock (two people: about 37 kg).
2. Flip the half over onto the room side: it opens on its hinges to lie flush with the top. Swing the gate leg down until it stops.
3. To fold: gate leg up flat, half back over onto the top, then lift the pair up to the wall and turn the buttons.



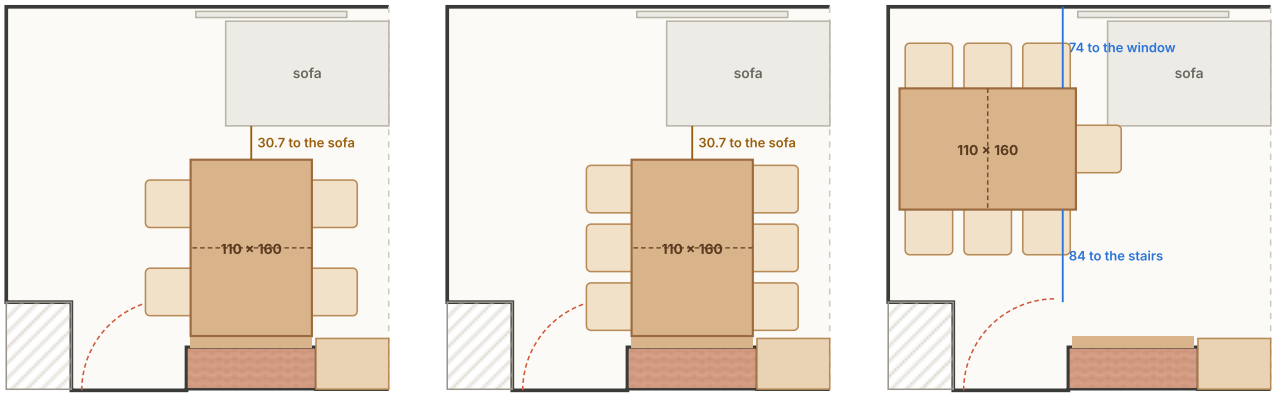
The flip-top table for four, six or seven

To scale, the west half of the room from above, window at the top. Red: the door's swing. Blue: the clearances that matter

Four, on the breast
the flip-top open: two down each side

Six, on the breast
three down each side, elbows in

Seven, by the stairs
three a side and one at the end



- Per diner: 60 cm of table edge (55 at a squeeze). Table edge to a wall: 60 cm is a tight minimum, 75 to aim for, 100+ to walk behind someone seated.
- The flip-top never leaves the wall; the dashed line is the joint where the half flips out on its gate leg. Red: the door's swing.
- Chairs drawn at IKEA TERJE size (43 × 49, seat 48): 28 cm from seat to table top, the usual 25–30.

Cut list for this step

Part	From	Size (cm)	Qty	Notes
Dining backboard · hinge rail	Solid oak, hardwax oil	110 × 22 × 3 · 110 × 4 × 7.3	1 · 1	From the same worktop's offcut

Table top and the half, centres	Solid oak, hardwax oil	101 × 80 × 2.7	2	Edge-glued staves: 96 between the shoulders, plus a 2.5 tongue each end (v15: the half matches the top)
Breadboard ends	Solid oak, hardwax oil	80 × 7 × 2.7	4	Pegged with three 10 mm pegs each, in slotted holes
Gate legs	Solid oak, hardwax oil	73.3 × 4.5, tapering to 3	2	Tapered on the two inside faces; tenoned into the apron (v15)
Gate apron	Solid oak, hardwax oil	94 × 7 × 2	1	Between the legs; two butt hinges along its top edge, under the half
Turn buttons	Solid oak, hardwax oil	6 × 2.5 × 2.2	2	On a single screw each
Häfele Hebgo folding brackets	Hardware	287.43.428, 580 arm	2	500 kg a pair
Lift-off hinges	Hardware	Union PowerLoad 100 × 88, all the same hand	3	
Flip-top hinges	Hardware	Lee Valley 40574, solid brass, 1.27 × 7.0	3	Routed into the meeting edges of the top and the half
Gate hinges · folding stays	Hardware	75 mm steel butt hinges · locking stays	2 · 2	The gate swings down and locks; folds flat inside the half

Order the oak

1. Ask the timber merchant for two centre panels as edge-glued staves, 101 × 80 × 2.7 each: 96 between the shoulders plus a 2.5 tongue at each end. Most will also cut the four breadboard ends and radius the half's two free corners.

The breadboard ends (both panels)

2. On each end of each centre panel, rout a tongue 2.5 long and 0.9 thick, centred in its thickness: take 0.9 off each face with the router against a batten.
3. In each breadboard end's inner edge, rout a groove 0.9 wide and 2.6 deep, centred, full length.
4. Dry-fit: the breadboard ends flush with the panel's long edges, flat across the joint.
5. Drill three 10 mm peg holes down through each breadboard end and the tongue in it: one at the centre and one 8 cm in from each long edge, 1.3 cm from the breadboard's inner edge.
6. Take the ends off. Elongate the two outer holes in each tongue to 16 mm long, along the panel's width: the panel moves with the seasons, the breadboard doesn't.
7. Glue only the middle 10 cm of each tongue and groove, clamp the ends on, and drive the pegs. Glue the outer pegs into the breadboard only, not the tongue. Saw flush and sand.

Shape and finish

8. Radius the half's two free corners to 5 cm (the top's corners meet the wall and the half, so they stay square): jigsaw outside the line, then sand to it.
9. Chamfer the underside edges 1 cm, except the two that meet at the joint, so it reads slim.
10. Sand both to 180. Oil both faces of both panels equally, every edge too: a panel oiled on one side cups.

The joint: three flip-top hinges

11. Lay the top and the half face up, their meeting edges together, flush across the top. Mark three hinges: one in the middle and one 15 cm in from each side.
12. Each hinge's two leaves set into the two edges; rout the recesses to the hinge's depth with a guide, square them with a chisel, and screw the hinges in. The pin sits so the half swings over onto the top through 180° and lies flat on it.
13. Swing it: open, the two tops are flush; folded, the half lies flat on the top with nothing proud.

The gate leg

14. Tenon the two legs into the ends of the apron, the legs 8 cm in from the half's sides, square. Glue and clamp.
15. Screw the two butt hinges along the apron's top edge and up into the half's underside, 3 cm in from its far edge, so the gate swings down to stand upright and up to lie flat along the half's 80 cm.
16. Fit a folding locking stay from each leg to the half's underside: upright, they lock; push the knuckle to fold.
17. Stand it on the floor upside down and check: folded, the legs lie flat with 1.5 cm to spare before the joint.

The backboard and hinge rail

18. First, knock along the stairs wall where the backboard goes (v17). A solid sound: brick, so plugs and screws as below. A hollow sound: plasterboard on studs. Then stop and find the studs with a detector before drilling anything; the screws go into the studs (or the fixings change), so tell us what you find.
19. Glue and screw the hinge rail along the backboard's top edge, flush at the back, so it stands 7.3 cm proud of the backboard's face.
20. Hold the backboard on its marks: centred on the wall, the rail's top at 73.3, level.
21. Drill through it and into the wall every 20 cm in two rows, 3 cm from its top and bottom edges. Plug, then fix with 5 × 70 countersunk screws.

The brackets

22. Screw the two Hebgo brackets to the backboard's face, 80 cm apart centre to centre (15 cm in from each end), their arm tops level with the rail's top.
23. Use long screws through the backboard into plugs in the wall, as many as the bracket's holes take: the 500 kg rating depends on the fixing.
24. Raise both arms until they click and lock. Lay a level across them.

Hang the table

25. Screw the three lift-off hinges along the rail's front edge, 10 cm from each end and one in the middle, pin halves on the rail.
26. Screw the other halves to the underside of the top's back edge (the long edge that meets the wall) at the same spacing, all the same hand.
27. With your helper, lower the top onto the pins and let it rest on the arms: at 76, level, in line with the alcove's worktop.
28. Flip the half out and lower its gate leg: it should stand level with the top all along. Adjust with felt glides if the floor isn't flat.

Folded

29. Fold the gate flat, flip the half back over onto the top, then lift the pair to vertical together: two people, one at each corner (about 37 kg).
30. Its top edge is at 153.3. Screw the two oak turn buttons to the wall just above it, 15 cm in from each end. Fit the magnetic catch behind the top edge.
31. Open and fold it three times: it shouldn't bind, both arms should click, the gate's stays should lock, and nothing should move when you lean on it open.

Check before moving on: It opens onto both brackets with a click each side, the half flips out flush and its gate leg locks upright; it folds back flat and swings up to the wall without binding, and nothing moves when you lean on it open.

Step 12 · Lighting: smart spot bars and the LED strips, your own job · You + a helper · 1 day

LIGHTING

Buy first: 7 Lighting (the buy list).

Parts

- Both Wolston-style spot bars and eight smart colour E14 bulbs under 6 W (WiZ Colour P45 4.9 W, or Hue E14 lustre 5.1 W)
- Smart colour strip, RGB and tunable white (about 4.5 m: Hue Lightstrip Plus or WiZ, in the bulbs' app), three aluminium channels with opal diffusers, its plug-in power
- 6500K bias light for the TV
- Your red lamp, with its colour bulbs

Hardware and fixings

- Connector block or lever connectors (if the bar's flex needs them)
- Earth sleeving, insulating tape

Tools

- Two-pole voltage tester (GS38) and its proving unit
- MCB lock-off and padlock
- Insulated screwdrivers, wire strippers
- Step ladder
- A helper to hold the bar

Before you start

1. No dimmers: smart bulbs need full power, so leave the light switches as they are and keep them on. A dimmer in front of a smart bulb makes it flicker and can damage it.
2. The same stop list as step 3. Here, also stop if a ceiling point has no earth (the bars need one), or the rose has more wires than you can match to your photo.

Isolate the lighting circuit

3. Tell everyone in the house you're working on the electrics.
4. At the consumer unit, switch off the breaker for the lighting circuit (usually marked "lights", 6 A). If the labels aren't clear, switch off the main switch instead.
5. Fit the MCB lock-off and padlock to that breaker, keep the key in your pocket, and tape a note on the board: "Working on this circuit, do not switch on".
6. Prove your voltage tester on its proving unit: it must light.
7. At each switch and each ceiling point, with the cover off, test between every pair: live and neutral, live and earth, neutral and earth. No reading on any of them.
8. Prove the tester on the proving unit again. Only now is it safe to touch.

Take down the old fittings, hang the bars

9. Unscrew each old fitting's cover. Prove dead again at its terminals. Photograph the wiring, mark the switched live (the wire from the switch) with tape, then disconnect and take the fitting down.
10. Fix bar 1's mounting plate to the ceiling over the rose between the TV and the sofa, the bar running from the TV towards the window (v15), with screws into the joist or the plate's fixings. Bar 2's goes on the pendant point by the stairs, running the same way, along the west wall.
11. Connect earth first (green-and-yellow to the bar's earth), then neutral to N and the switched live to L, in the bar's terminal block; any loop wires go back together exactly as they were. Tighten, tuck, and fit the canopy.
12. Fit the smart colour bulbs and point every head straight down, evenly spread along the bar (v17). Tilt one only if its pool lands in the TV's screen or someone's eyes. If a WiZ bulb shows below its shade's rim, use the shorter Hue lustre there.

Power on and test

13. Remove the lock-off and switch on, wall switches on. In the bulbs' app, add all eight, group them as bar 1 and bar 2, set warm white at about half, then save the five scenes (General first) from the lighting scenes below (levels, and colours if you want them). Your red lamp's colour bulbs go in the same scenes.

The low-voltage lights (nothing to isolate)

14. Cut the smart strip at its marks into three lengths: one for each upright's groove and one for the side-wall run. Press each into its aluminium channel under the opal diffuser; take the uprights' leads down their grooves to the strip's power on the 10-gang (v16: smart colour).
15. The side-wall run's strip: in its channel in the groove under the run's front edge, 1.5 cm back from the edge, the full 220 cm; its lead drops with the PC's through the brush grommet into the conduit, to the same power.

16. Add the strips to the app beside the bars' bulbs, so each scene sets the wall's glow too: warm white for everyday, low and coloured for TV night.
17. Clip the ceiling strip through its notches in the uprights; stick the bias light round the back of the TV on its USB port.
18. Stand your red lamp in its spot (the 3D room shows the three), its bulbs green, pink and purple, top to bottom.

Check before moving on: Every bulb answers the app, the scenes recall, and from the sofa no spot head shows in the screen or shines in your eyes.

Step 13 · Connect and finish · You · 1 day

AV NETWORK

Buy first: 8 TV, consoles and AV, 9 Vinyl kit, 10 Finishing (the buy list).

Parts

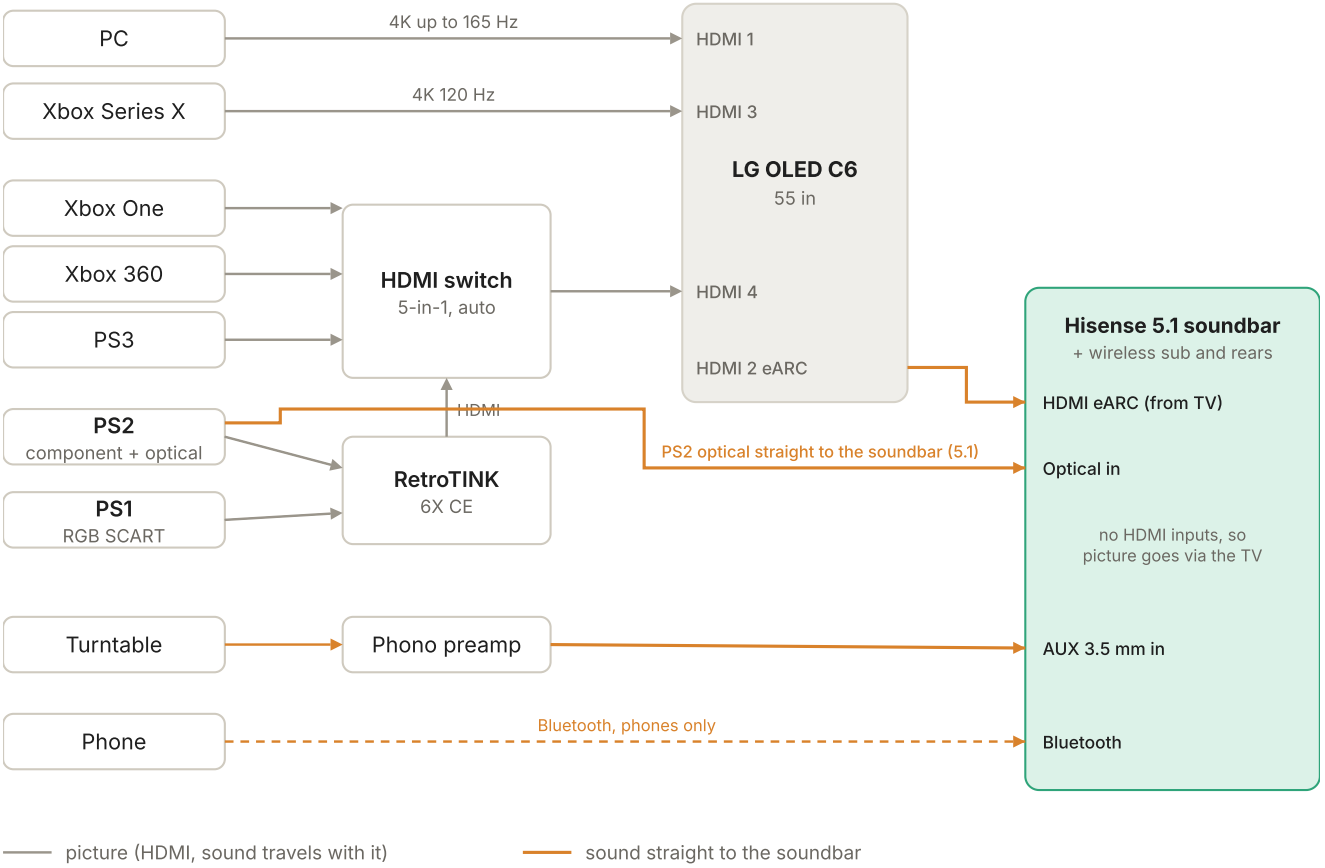
- HDMI switch, RetroTINK, HDMI leads, optical and SCART cables, the active optical HDMI for the PC
- Stylus, preamp, RCA-to-3.5 mm lead, record brush
- Labels

Tools

- Phone with the Deco app
- Velcro ties

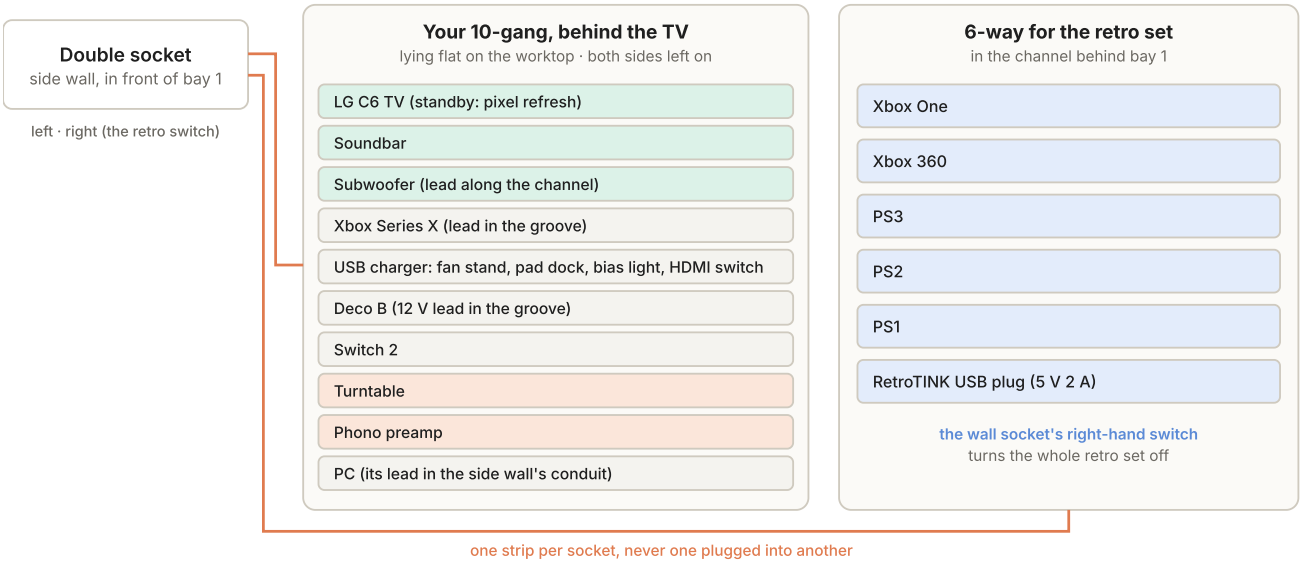
Everything sounds through the Hisense: directly where it can, via the TV where it can't

The soundbar has no HDMI inputs, so HDMI sources reach it over eARC; vinyl, PS2 audio and phones go straight in



Power map: your 10-gang (now with the PC) plus a 6-way for the retro set

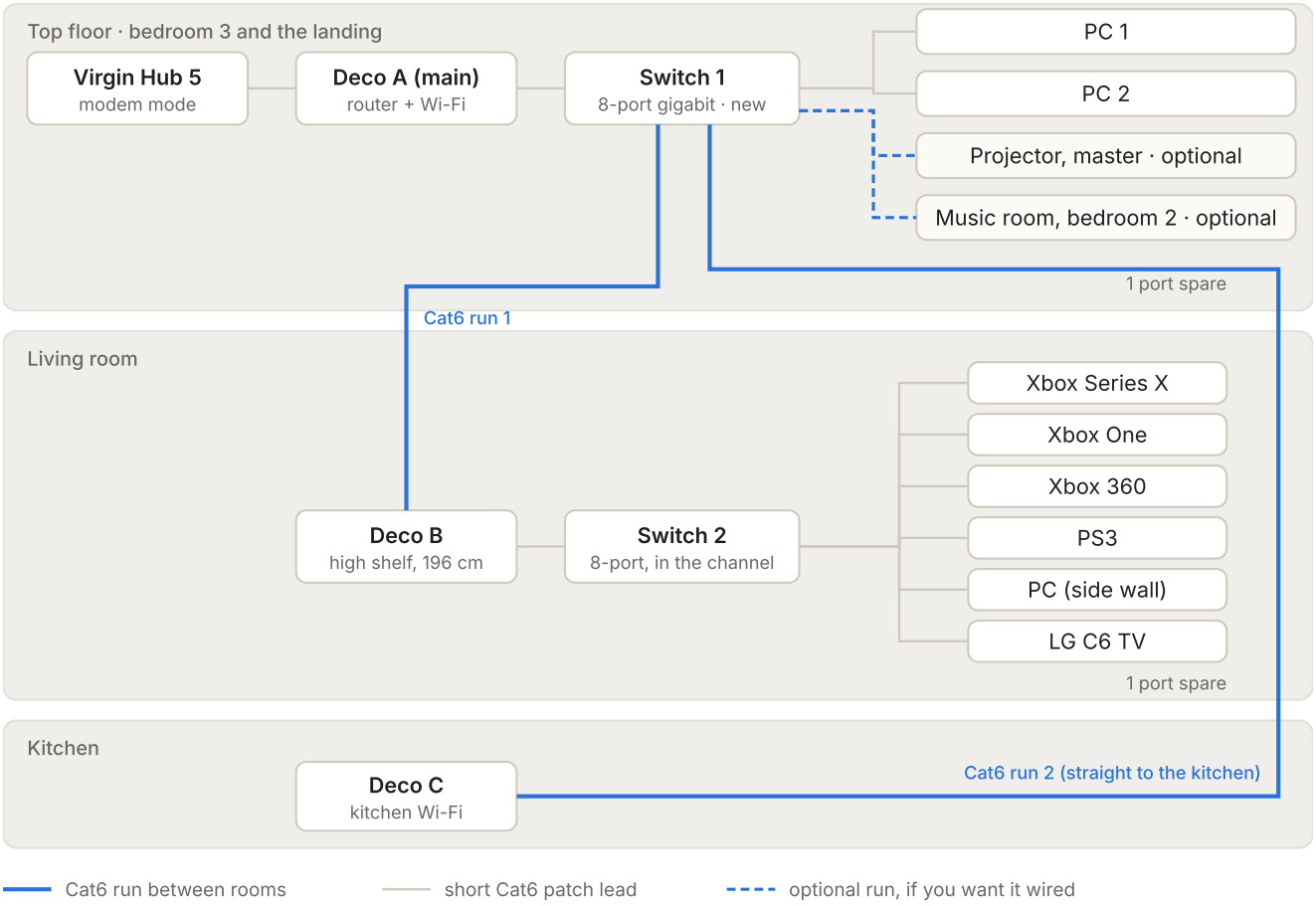
Everything modern on the left outlet, always on; the five retro consoles and the RetroTINK on the right, switched at the wall



- The socket on the side wall is a single today: you swap it for a double at the same spot (like for like, your own job: step 3).
- The retro switch moves to the wall: with the TV 2 cm from the left upright, the 10-gang's own switches can't be reached any more.
- Rear speakers: your double socket behind the sofa's short side (v23). Your lamp and the grow lights: the double behind the plant stand.
- The sub plugs into the 10-gang: its lead drops through the worktop's slot to the back of bay 2. Both spot bars use existing ceiling points.
- Load check: the 10-gang about 1.55 kW with everything on, PC included (850 W at most); the 6-way up to 0.8 kW. Each under 13 A (3 kW).
- Check the 10-gang is surge-protected; if not, use a surge-protected one for the TV and consoles. Label both outlets.

Every floor has its own Cat6 link back to bedroom 3

Deco A and Switch 1 live with the Hub 5 and the gaming PCs in bedroom 3; two spare ports can wire the projector and the music room



1. TV inputs: HDMI 1 the PC, HDMI 2 the soundbar (eARC), HDMI 3 the Series X, HDMI 4 the HDMI switch (Xbox One, 360, PS3, RetroTINK).
2. The PS2 by component into the RetroTINK and by optical to the soundbar; the PS1 by RGB SCART.
3. The deck to the preamp, the preamp to the soundbar's AUX; set the tracking weight; soundbar in Music mode for vinyl.
4. Network stage 2: plug each Cat6 run into its Deco, check Deco B and C show "Ethernet", and give the Series X and PC high priority.
5. Label both outlets of the double socket and each HDMI switch input; set the sub and rear levels from the sofa.

Check before moving on: Everything on the network's "done when" list, and every source plays through the soundbar.

Shopping list

About £3,223–6,463, in build order; no electrician on top.

Status	Item	Qty	Estimate (£)	Paid (£)
NETWORK **1 NETWORK, NOW**				
☐ To do	Put the Virgin Hub 5 into modem mode			
☐ To do	Set up Deco A, Switch 1 and the PCs in bedroom 3			
☐ To do	Add Deco B and Deco C over Wi-Fi for now; speed-test each room and the Xbox			
☐ To buy	Cat6 patch leads, 0.5–2 m	11	15–20	
☐ To buy	TP-Link TL-SG108 8-port switch: one for the living room, one to replace the 5-port in bedroom 3	2	36–50	
POWER & CABLES NETWORK **2 CABLES AND POWER**				
☐ To do	Cat6, your own job: two runs from bedroom 3, across the landing and down through the airing cupboard, one to the alcove's right corner (Deco B) and one on to the kitchen (Deco C), a spare pulled with each, and the optional runs to the master and bedroom 2 **(changed: no electrician)**			
☐ To buy	Cat6 solid pure copper, round (not CCA): two runs, four with the optional ones	2–4 runs	30–60	
☐ To buy	Cat6 tools: a punch-down tool, RJ45 keystones and two surface boxes (or plugs and a crimper), a cable tester, a draw-rod set **(new: you run the Cat6 yourselves)**	1 set	30–60	
☐ To do	Swap the single socket on the side wall for a double at the same spot: like for like, not notifiable; step 3 has the safe way to do it **(changed: your own job)**			
☐ To buy	A double socket in the same finish, and a deeper back box if the old one is shallow **(new)**	1	5–15	
☐ To buy	Electrical safety kit: a two-pole voltage tester to GS38 with its proving unit, an MCB lock-off and padlock, a plug-in socket tester **(new: for doing it yourselves)**	1 set	45–90	
☐ To buy	Insulated (VDE) screwdrivers, wire strippers and side cutters **(new)**	1 set	20–40	
☐ To buy	6-way surge-protected extension for the retro set (Xbox One, 360, PS3, PS2, PS1, RetroTINK) **(changed: switched at the wall)**	1	20–35	
☐ To buy	4-port USB charger, 5 V 2 A per port (HDMI switch, fan stand, pad dock, bias light)	1	10–20	
☐ To buy	USB plug, 5 V 2 A or more, for the RetroTINK on the retro 6-way	1	5–10	
☐ To buy	Cable kit: brush grommets, velcro ties, labels	1	20–30	
☐ To buy	Flat two-compartment trunking for the 4 cm channel behind the base, about 2 m	1	10–20	
☐ To do	Chase two conduits into the solid side wall for the PC, from behind the PC shelf down to skirting height and behind the skirting to the base's plinth: 38 mm oval for the HDMI and Cat6, 25 mm oval for the PC's mains lead; make good and paint **(changed: your own job; the PC now runs from the 10-gang)**			
☐ To buy	Conduit: 38 mm and 25 mm oval, about 4 m each, two draw wires, one oak-coloured brush grommet; a 5 m C13 lead with a rewireable 13 A plug for the PC **(changed)**	1	15–35	
☐ To buy	For the chase: a wall chaser with dust extraction for a day (or a bolster chisel and club hammer), one-coat filler **(new)**	1	20–45	
☐ To buy	PC tidy kit: braided sleeve, shelf-notch grommet	1	5–15	
☐ To buy	A powered USB 3 hub, 4-port with its own adapter, screwed under the side run's front edge below the PC: pads, the keyboard and a stick within reach of the sofa **(new in v17)**	1	15–30	
☐ To buy	A remote power button for the PC: a wireless switch on the motherboard's power header (card in a spare slot), its push button on the wrap ledge **(new in v17)**	1	10–30	
☐ To buy	A magnetic mesh dust filter cut to the PC's top exhaust **(new in v17)**	1	5–10	
PAINT **3 PAINT**				
☐ To do	Sample Farrow & Ball Card Room Green No. 79 on an A4 board; check it by day and with the LED strip on			
☐ To buy	Card Room Green: Estate (or Modern) Emulsion for walls, ceiling, alcove and chimney breast; Estate Eggshell for skirting and radiator (the door and its frame stay white, v20); primer and undercoat		200–500	
☐ To buy	Woodwork primer, sanding pads and sugar soap for the white skirting and door	1	20–40	

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To do	Paint the whole room before anything is fixed: alcove and breast included; skirting and door from white to eggshell			
● MEDIA WALL	**4 BASE UNIT**			
☑ Done	Check the alcove back wall for fixings (solid)			
☐ To do	Measure the alcove's width at the floor, 76, 134, 155 and 196 and the ceiling, and the floor's level across it			
☑ Done	Check the side wall where the ledge and PC shelf go, and the chimney breast where the dining wall goes: solid (you confirmed) **(new)**			
☐ To buy	Two IKEA BESTÅ 60×40×64 frames, white stained oak effect, for bays 1–2	2	70–200	
☐ To buy	Plinth for bays 1–2: 18 mm ply ladder frame 120 × 38 × 9 cm with a clear bay under bay 2, plus shims	1	15–30	
☐ To buy	Hidden plinth drawer: oak front 54 × 6 with a finger-pull groove underneath, 12 mm ply box 52 × 34 × 5, 300 mm full-extension runners **(changed: no push-to-open at floor level, as v6.1 said)**	1	30–50	
☐ To buy	Bay 3 carcass: 18 mm birch ply, 60 × 40 × 73 cm from the floor, connector bolts, and a 12 mm birch-ply divider 38 × 69.4 that splits it into the slim game column and the LP column **(changed in v16: the divider)**	1	50–90	
☐ To buy	Two face-out LP crates, 34.9 × 40 (a 33 cm trough): 12 and 9 mm birch ply cut to size, 15 mm oak lips, raked rests, felt, two pairs of 400 mm runners rated 68–70 kg **(changed in v16: snug crates, no dead space)**	2	120–200	
☐ To buy	Three slim drawers for the game column, 15.3 × 40: 6 mm birch ply sides and backs, 12 mm bases, oak lips, felt, three pairs of 400 mm runners (light duty is enough); a 2 m USB-C lead and a grommet for the pads drawer **(new in v16)**	3	70–130	
☐ To buy	BESTÅ shelves, 56 × 36 cm (20 kg max each), two each for bays 1 and 2	4	12–32	
☐ To buy	The sub in bay 2: an isolation riser for it (dense foam or sorbothane, about 22 × 28 × 3), 12 mm birch ply for the divider (35.3 × 38) and the part shelf (32.2 × 36), four shelf pins, felt pads **(new in v13.5: the sub moves into the media wall)**	1	25–45	
☐ To buy	Anti-tip brackets and 4 cm spacer blocks: both frames and bay 3	1 set	15–25	
☐ To do	Measure about ten games of each kind (PS2, Xbox 360, Xbox One) before cutting the game drawer			
☐ To buy	Game drawer kit: 12 mm birch ply cut to size, 350 mm full-extension soft-close runners (35–49 kg), oak lip, follower block, felt	1	50–90	
☐ To do	Build the plinth and bay 3, fix everything to the wall, fit the shelves, drawers and trays; place the TV and soundbar			
● MEDIA WALL	**5 OAK JOINERY**			
☐ To buy	Two solid oak uprights, 165 × 30 × 4.5 cm (cut to 159), quarter-sawn if possible	2	150–300	
☐ To buy	Solid oak for the shelves: high shelf in two lengths (128.9 and 51.7 × 30 × 4), Series X shelf 55 × 30 × 4, record ledge 128.9 × 7 × 2.4, the side-wall run from one board 225 × 30 × 4 (cut to its sweep), lips **(changed in v15: one run for the ledge and the PC shelf)**	1 set	230–380	
☐ To buy	Oak wedges, 10 mm oak dowel for pegs and dowels, concealed upright brackets; two concealed shelf brackets for the PC shelf, rated 40 kg+ a pair	1	30–60	
☐ To buy	One solid oak worktop, 300 × 62 cm, full stave: the alcove top (186 × 46) and the dining wall's backboard (110 × 22) from the offcut, chamfered fronts **(changed)**	1	250–450	
☐ To buy	Oak face frame, 20 mm, 186 cm for the base	1	60–120	
☐ To buy	Keyboard tray for the PC shelf: 7 cm oak housing under it, 12 mm ply tray 54 × 26 with an oak lip, 300 mm light-duty runners **(new: back from v6.1)**	1	25–45	
☐ To do	Cut the joints and fit the uprights, shelves, ledges and face frame; notch the uprights for the ceiling strip			
● DINING TABLE	**6 DINING WALL**			

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To buy	Solid oak for the flip-top: two panels 110 × 80 × 2.7 with pegged breadboard ends, the top and the half (edge-glued staves from a timber merchant, who can also radius the corners) **(changed in v15: the half replaces the leaf)**	1 set	280–480	
☐ To buy	Two Häfele Hebgo heavy-duty folding brackets, 287.43.428 (580 arm, 500 kg a pair) **(new)**	2	70–120	
☐ To buy	Union PowerLoad lift-off hinges, 100 × 88, one 3-pack (all the same hand) **(new)**	1	16–20	
☐ To buy	Three Lee Valley 40574 narrow flip-top hinges (solid brass, 1.27 × 7.0 cm) for the joint **(new in v15)**	3	25–45	
☐ To buy	The gate leg: two tapered oak legs 73.3 × 4.5 to 3 and a 94 × 7 × 2 oak apron (from the cutting list), two 75 mm butt hinges, two folding locking stays **(new in v15: instead of eight legs and the leaf's leg case)**	1	40–90	
☐ To buy	Oak for the hinge rail (110 × 4 × 7.3), two oak turn buttons, a magnetic catch, felt pads **(changed in v15)**	1	15–30	
☐ To buy	Folding chairs, IKEA FRÖSVI (TERJE's successor, the same size) or similar with a hole in the back for a hook, four, or six for six at the table (skip if yours fit: seat about 45–48) **(changed in v20)**	4–6	100–240	
☐ To buy	The chair cupboard beside the table: oak (or oak-faced ply) for its two sides, top, floor and shelf and two doors (cutting list), a 6 mm back, four concealed soft-close hinges, two push latches, two screws and plugs for its top rail **(new in v20)**	1	120–240	
☐ To buy	Two LP frames for the chimney breast, hung from 165 cm	2	20–40	
☐ To buy	The “now playing” ledge on the breast, with the table by the stairs: oak 105.5 × 7 × 2.4 with a 1.5 lip (from the shelf board's offcut), two concealed brackets **(new in v16)**	1	10–25	
☐ To do	Fix the backboard and hinge rail, hang the brackets and the top, fit the half on its flip-top hinges and the gate leg under it; check it opens, locks, folds and folds up			
☐ To do	Build the chair cupboard and stand it in the gap between the table's window end and the pipe boxing: level it, screw its top rail to the wall, hang the doors to swing towards the table; stand two folded chairs on its floor and two on its shelf **(new in v20)**			
LIGHTING **7 LIGHTING**				
☐ To do	Swap both ceiling fittings for the spot bars, turned to run from the TV towards the window: like for like, your own job; step 12 has the safe way. The light switches stay as they are (smart bulbs need full power) **(changed in v15: no dimmers)**			
☐ To buy	Eight smart colour E14 bulbs for the bars: WiZ Colour P45 E14, 4.9 W (Wi-Fi, no hub), or Philips Hue White and Colour E14 lustre, 5.1 W, if a WiZ shows below its shade; both under the bar's 6 W **(new in v15)**	8	100–360	
☐ To buy	Smart colour strips for the upright grooves and under the side-wall run, in the same app as the bulbs: about 4.5 m of RGB and tunable-white strip (Philips Hue Lightstrip Plus, or WiZ's strip kits, cut at their marks), three aluminium channels with opal diffusers, plug-in power on the 10-gang (no fused spur); a 6500K bias light for the TV stays plain white **(changed in v16: smart colour, so the whole wall changes with a scene)**	1	90–220	
☐ To buy	Two Dunelm Elements Wolston 4-light spotlight bars, £49 each (bulbs above) **(changed: two)**	2	98–110	
☐ To buy	Optional: a smart button for the bars, so a scene is one press rather than the app (WiZ remote or Hue dimmer switch) **(new in v15)**	1	0–25	
☐ To do	Set up the smart bulbs in their app: group them as bar 1 and bar 2, then save the five scenes (General first; Paint and light) with their levels and colours; your red lamp's bulbs go in the same scenes **(new in v15)**			
☐ To buy	Later: aluminium channel and opal diffuser for the existing ceiling LED strip (optional)	1	0–80	
AV **8 TV, CONSOLES AND AV**				
☐ To buy	5-in-1 4K60 HDMI switch with remote	1	20–35	
☐ To do	Measure the PC-to-TV cable route (about 4–5 m along the skirting)			

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To buy	Active optical HDMI cable for the PC, certified Ultra High Speed (48 Gbps), about 5 m	1	20–60	
☐ To buy	High Speed HDMI leads, 1–2 m; two 90° HDMI adapters in case the TV's plugs need them	5 + 2	25–40	
☐ To buy	RetroTINK-6X CE upscaler for the PS2 and PS1 (order on restock; UK price includes VAT and import fees)	1	200–240	
☐ To buy	PS2 component video cable, good quality	1	10–20	
☐ To buy	Optical (TOSLINK) cable, 2 m, PS2 straight to the soundbar	1	5–10	
☐ To buy	RGB SCART cable for the PS1	1	8–15	
☐ To do	Connect the TV inputs; set up the PS2 and RetroTINK			
AV **9 VINYL KIT**				
☐ To buy	Replacement stylus for the turntable (exact match for its model)	1	15–40	
☐ To buy	External phono preamp (only if the deck has a PHONO/LINE switch)	1	40–100	
☐ To buy	RCA-to-3.5 mm lead, 2 m: preamp under the Series X shelf to the soundbar's AUX **(changed: shorter)**	1	5–10	
☐ To buy	Isolation feet for the turntable; a carbon-fibre record brush	1	20–40	
☐ To do	Set the tracking weight; soundbar in Music mode			
AV NETWORK **10 FINISHING**				
☐ To do	Turn on HDMI-CEC (SIMPLINK); PS3 Control for HDMI off			
☐ To do	Deco app: prioritise the Series X and PC, guest Wi-Fi, automatic firmware			
☐ To do	Label both outlets of the double socket (left: everything, right: retro) and each HDMI switch input			
☐ To do	Set the sub and rear speaker levels from the sofa			
☐ To buy	Drip trays or a waterproof mat under the pots on the plant stand's slatted tiers; if you add clip-on grow lights later, a smart plug with a schedule **(new in v18)**	1	10–20	
☐ To buy	Felt pads under the vinyl stand's feet, so it slides out from the breast without marking the floor **(new in v18)**	1	3–6	
☐ To buy	A rug about 230 × 160 in front of the sofa: cream with deep blue, rust and ochre goes with the green, the oak and the blue sofa (the plan draws one as a guide) **(new in v15)**	1	120–300	